

# Minute-scale single-cell transcriptomics enables dynamic modeling of cellular behavior

Kanishk Asthana<sup>†‡,1,2</sup>, Alexander N. Jambor<sup>†,1,2</sup>, and Wei Wang<sup>‡,1,2,3,4</sup>

*1 Department of Chemistry and Biochemistry, University of California San Diego, La Jolla, CA 92093, USA*

*2 Bioinformatics and Systems Biology PhD Program, University of California San Diego, La Jolla, CA 92093, USA*

*3 Department of Cellular and Molecular Medicine, University of California San Diego, La Jolla, CA 92093, USA*

*4 Lead contact*

*† Equal contribution*

*‡ Corresponding authors*

*Corresponding author contact information:*

*Kanishk Asthana: [kasthana@ucsd.edu](mailto:kasthana@ucsd.edu)*

*Wei Wang: [wei-wang@ucsd.edu](mailto:wei-wang@ucsd.edu)*

## Abstract

Dynamic cellular processes such as signaling, fate decisions, and intercellular communication unfold on minute timescales, a regime inaccessible to conventional transcriptomic methods. This temporal gap has fundamentally limited the development of predictive, causal models of cell behavior. Here, we bridge this gap by introducing ChronoSeq, an automated single-cell RNA sequencing platform that achieves genome-wide profiling with a temporal resolution as brief as seven minutes. By integrating automated live-cell sampling with molecular time-barcoding, ChronoSeq captures rapid transcriptional dynamics with high fidelity. Applying ChronoSeq to TNF- $\alpha$  stimulated cells, we discovered a rapid, heterogeneity-driven bifurcation in the NF- $\kappa$ B response that was previously unobservable. We further demonstrate that the high-density temporal data generated by ChronoSeq enables a new class of computational models that dramatically outperform existing methods in inferring the directionality and targets of post-translationally regulated transcription factors. Finally, in a multicellular co-culture, ChronoSeq resolved a paracrine signaling cascade in real time, identifying both the timing and molecular identity of the intercellular relay. By providing a framework to measure dynamics, infer regulation, and model communication at the true pace of biology, ChronoSeq establishes a new foundation for dynamic systems biology.

## Introduction

Building predictive, dynamic models of cellular behavior, the vision of a “Virtual Cell”<sup>1,2</sup>, requires experimental data that reveal how gene regulation and cell-cell communication unfold in real time. Achieving this vision depends on three core capabilities: capturing transcriptional dynamics on the natural timescales of signaling and fate decisions, inferring causal rather than correlative gene regulatory networks (GRNs), and resolving the timing of multicellular communication that coordinates collective responses. Yet current methodological and data limitations have kept these goals largely out of reach.

Many cellular processes operate on minute timescales<sup>3–5</sup>, a regime inaccessible to standard single-cell RNA sequencing (scRNAseq)<sup>6–16</sup>. For example, the NF- $\kappa$ B signaling cascade activates transcriptional responses within 5 minutes of TNF- $\alpha$  stimulation and peaks at around 25 minutes<sup>17</sup>, timescales that are effectively invisible at the 30–60 minute resolution of existing scRNA-seq. Without dense, accurately timed expression measurements, computational efforts to infer causality or simulate cellular decision-making remain fundamentally constrained.

Several specialized methods have attempted to address this gap, but each fall short of combining fidelity, temporal resolution, and throughput. Metabolic labeling<sup>18–23</sup> and scGROseq<sup>24</sup> can detect nascent transcripts yet suffer from manual processing errors and low scalability. Live-seq<sup>6</sup> preserves cell viability but has hourly sampling intervals and limited throughput. Live-cell imaging<sup>25–27</sup> captures real-time trajectories only for a few marker genes and is hindered by phototoxicity and narrow genomic coverage<sup>28</sup>. No current technology achieves genome-wide, minute-scale measurements under physiological conditions with full experimental automation, capabilities that are essential for generating the benchmark datasets required for dynamic *in silico* modeling.

To bridge this gap, we introduce ChronoSeq, an automated droplet-based single-cell RNA sequencing platform that performs genome-wide profiling with a sampling interval as brief as seven minutes. ChronoSeq integrates automated sampling with molecular time-barcoding into a single workflow, preserving cell health while capturing transcriptional dynamics over extended perturbations. Together with a companion computational suite, ChronoPack, this technology provides an integrated framework to measure dynamics, infer causality, and model intercellular communication, the core components of Virtual Cell.

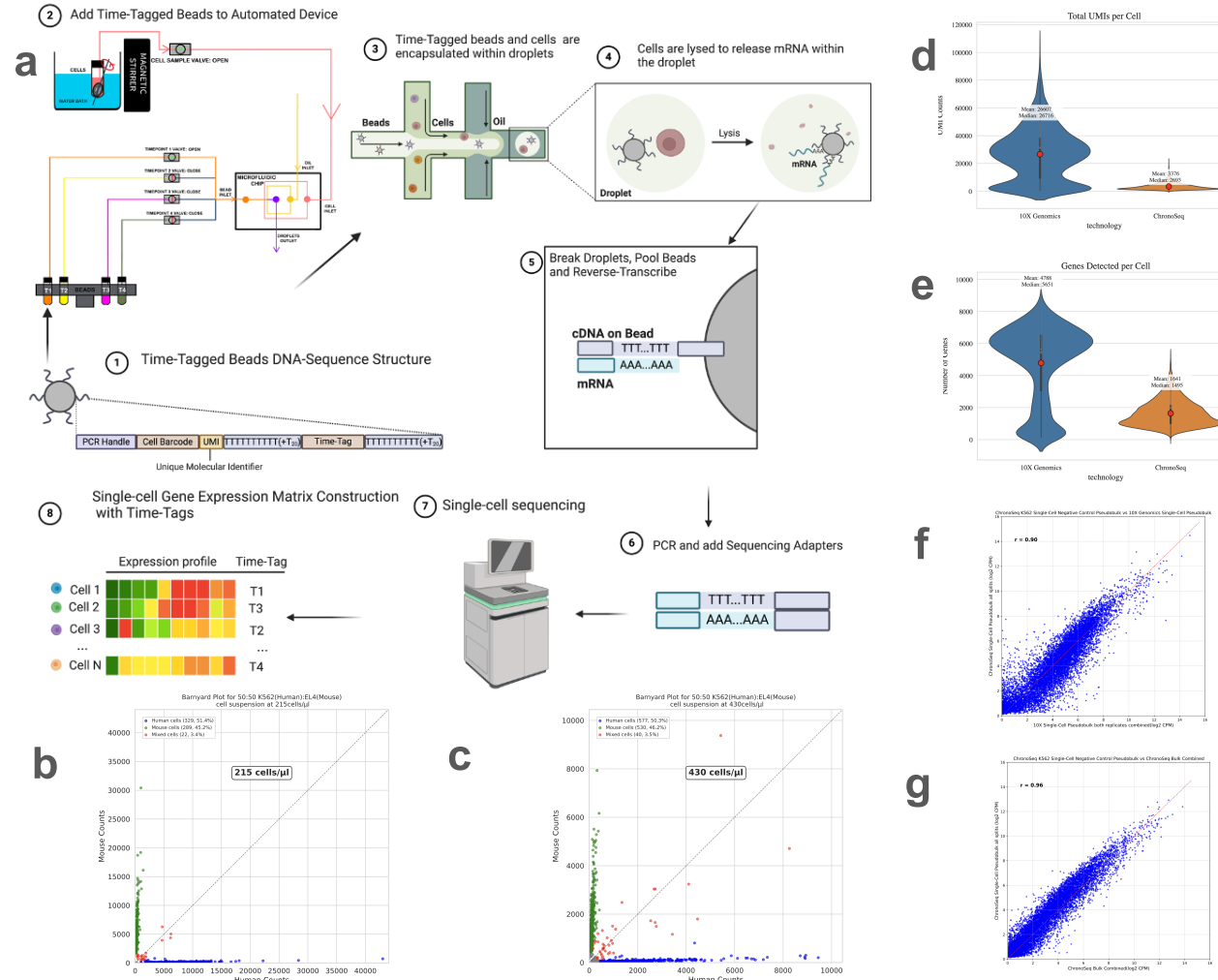
We demonstrate these capabilities through proof-of-concept studies addressing rapid signaling, causal inference, and multicellular communication. Minute-scale profiling of TNF- $\alpha$  stimulated K562 cells revealed a rapid bifurcation into erythroid and non-erythroid trajectories, directly capturing fate transitions as they occurred. Using ChronoPack, we reconstructed temporally informed GRNs that outperformed existing inference methods, yielding mechanistic insight into early transcriptional regulation. Finally, in a co-culture of THP1 macrophages and K562 cells, ChronoSeq resolved secondary paracrine signaling induced by the LPS stimulation of THP1 cells, pinpointing the timing and molecular identity of secreted factors mediating cross-cell activation.

By combining genome-wide, minute-scale temporal profiling with causal network inference and intercellular dynamics, ChronoSeq establishes a new foundation for dynamic systems biology. This integration brings the Virtual Cell concept within reach, linking molecular mechanism to phenotype in predictive, quantitative terms.

## Results

### An automated platform for minute-scale temporal scRNA-seq

To directly observe rapid transcriptional dynamics, we developed ChronoSeq, an automated platform that integrates live-cell culture with droplet-based scRNA-seq. The system performs programmed sampling at intervals as short as seven minutes, co-encapsulating each cell with uniquely barcoded Time-Tag beads (**Fig. 1a, Fig. S2, Supplementary Movie 1**). Immediate lysis upon droplet formation preserves the transcriptional state, while all timepoints are processed within a single library to dramatically reduce labor while eliminating batch effects and manual variability (**Fig. S1**). Continuous cell maintenance under physiological conditions ensures that captured profiles reflect authentic biological responses.



**Figure 1. (a) ChronoSeq Library Preparation Workflow:** Time-Tagged beads are added to reservoirs inside the ChronoSeq device (Step 1-2). Our current system supports 12 unique Time-Tags, each Time-Tag is added to its own dedicated reservoir. Cells suspended in a culture media are loaded into another reservoir maintained at 37°C and 5%CO<sub>2</sub>. The ChronoSeq device then automatically co-injects Time-Tags 1-12 with samples 1-12 from the Cell reservoir into a microfluidic chip at programmed time intervals of 7 minutes or longer (Supplementary Movie 1 and Figure S2). This results in droplets where cells collected from each sample 1-12 are present with ChronoSeq beads with the same Time-Tags 1-12 (Step 3). Cells are immediately lysed upon droplet formation and kept on ice to preserve the transcriptome until all 12 samples are collected (Step 4). For samples with high RNAase activity, RNAase inhibitor is added to each sample to help further maintain RNA integrity. Next, following the same process as Drop-seq, droplets are broken, pooled, and reverse transcribed together in unified library preparation workflow for all samples. The cDNA on the beads is amplified and sequencing adapters are added (Steps 5-6). The library is sequenced and processed with our ChronoSeq-Tools pipeline to create Digital Gene Express Matrices (DGEs) with a Time-Tag number for each cell (Steps 7-8). **(b,c)** Species-mixing barnyard plots for 50:50 K562(Human):EL4(Mouse) cells at two commonly used ChronoSeq working concentrations (215 cells/μl and 430 cells/μl). **(d-g)** We generated K562 negative control datasets using both single-cell and bulk sampling using the ChronoSeq device. For both bulk and single-cell datasets, K562 cells were maintained inside the ChronoSeq device and samples were taken using the ChronoSeq device at programmed intervals of 10 minutes. For the bulk dataset, cells sampled by the device

were manually mixed with Time-Tags 1-12. d-e) We downloaded both replicates of a published K562 dataset<sup>29</sup> generated using the commercially available 10X Genomics Chromium 3' single-cell RNA-seq technology for comparison: (d) shows a violin plot comparing mean and median transcripts captured, (e) shows a violin plot comparing genes detected for the 10X Genomics dataset and ChronoSeq single-cell K562 negative control dataset. (f-g) We prepared pseudobulks across all Time-Tags for both bulk and single-cell ChronoSeq K562 negative control datasets. We also prepared a combined pseudobulk for both 10X Genomics K562 replicates. (f) shows a correlation scatterplot between the 10X Genomics and ChronoSeq single-cell K562 pseudobulks. ChronoSeq shows a high concordance with 10X Genomics with a correlation coefficient  $r=0.90$ . (g) Comparison of ChronoSeq single-cell and bulk shows even higher concordance with a correlation coefficient  $r=0.96$ .

We manufactured (**Fig. S3**) and validated the fidelity of the 12-plex Time-Tag barcode system using mixed human–mouse samples. Each tag cleanly partitioned species-specific transcripts, with minimal barcode cross-talk ( $\leq 1.6\%$  mixed beads per tag; **Fig. S4**). Automated sampling within the ChronoSeq device showed similarly precise tagging and each cell population was exclusively labeled with its designated barcode (**Fig. S5**). ChronoSeq's single-cell capture efficiency matched commercial standards, with a doublet rate below 4% across operational cell concentrations (**Fig. 1b,c**). We captured an average of 636 cells per Time-Tag. Together, these results demonstrate that ChronoSeq achieves fully automated, high-integrity sample tagging and single-cell encapsulation.

To benchmark transcriptional accuracy independently of temporal resolution, we profiled unstimulated K562 cells using ChronoSeq. Despite the number of detected genes and transcripts per cell being lower than 10X Chromium (**Fig. 1d,e**), ChronoSeq exhibited quantitative fidelity: pseudobulk profiles from ChronoSeq single cells correlated nearly perfectly with ChronoSeq bulk RNA-seq ( $r = 0.96$ ) and strongly with 10x Chromium pseudobulks ( $r = 0.90$ ) (**Fig. 1f,g**). These results confirm that ChronoSeq accurately preserves transcriptional signatures, with minimal bias introduced by automated sampling. The high concordance establishes a robust foundation for interpreting minute-scale biological dynamics.

### **ChronoSeq reveals rapid fate bifurcation during the NF- $\kappa$ B inflammatory response**

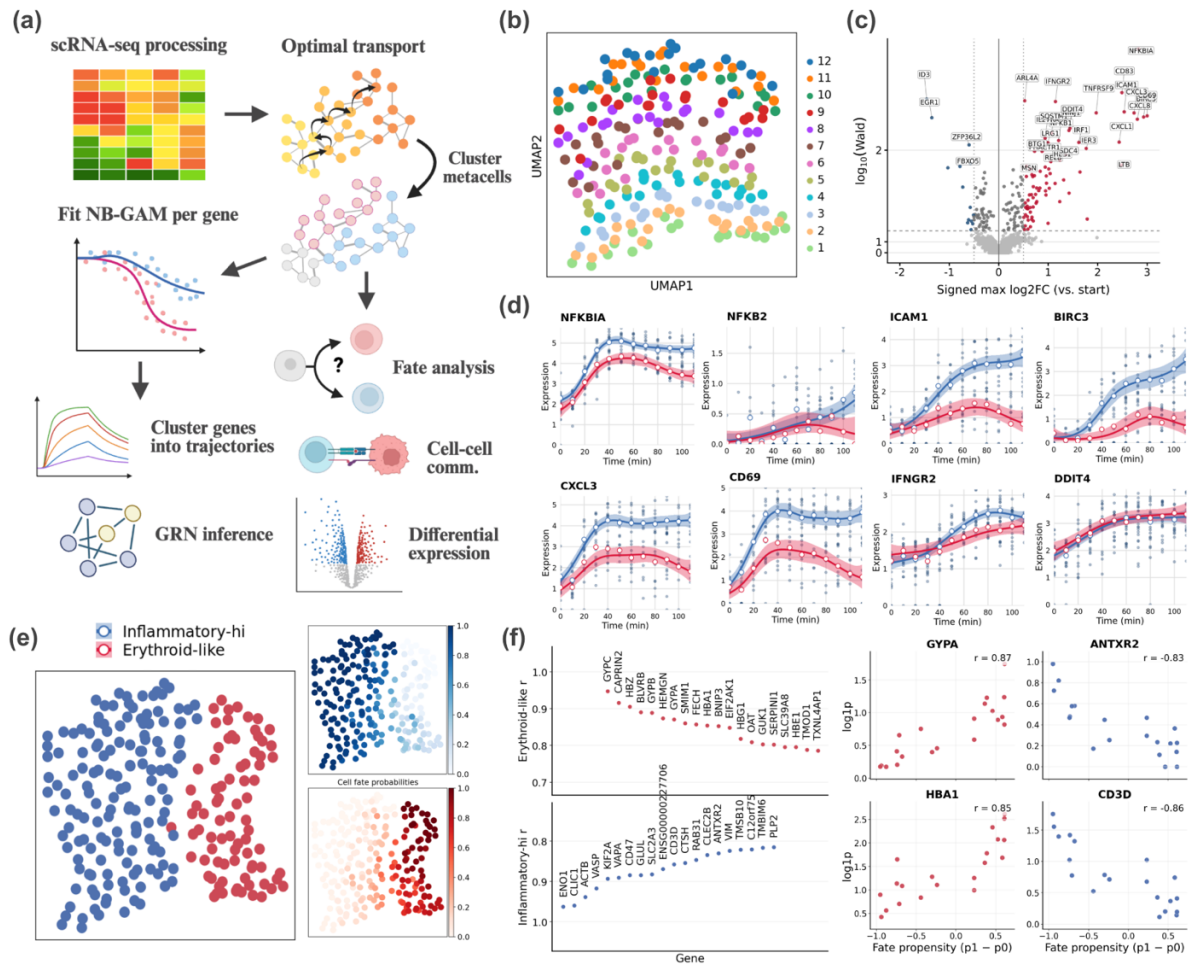
To demonstrate ChronoSeq's ability to resolve rapid transcriptional dynamics, we profiled K562 cells responding to TNF- $\alpha$  stimulation at 10-minute intervals over 2 hours, which is known to activate the NF- $\kappa$ B pathway. Temporal pseudobulk profiles showed high concordance with matched qPCR timecourses for representative NF- $\kappa$ B target genes (**Fig. S6**), validating ChronoSeq's ability to capture real-time dynamics.

We developed ChronoPack, an open-source computational suite for quantitative analysis of time-resolved single-cell data (**Fig. 2a**). The suite integrates standard preprocessing, expression correction, and metacell construction, followed by trajectory inference using WaddingtonOT<sup>30</sup> to derive a probabilistic cell-state transition kernel across time. This kernel enables cell-fate mapping, dynamic clustering, and the identification of early predictive genes. Temporal gene expression is modeled with negative-binomial generalized additive models (NB-GAMs) using tradeSeq<sup>31</sup> to detect statistically significant expression trends. Building on these results, ChronoPack can also infer causal gene regulatory networks from time-resolved dynamics and reconstruct paracrine signaling by integrating ligand–receptor databases with Granger causality analysis. Collectively, ChronoPack provides a unified framework for inferring causal regulation and intercellular communication from high-temporal-resolution scRNA-seq data.

Trajectory reconstruction using ChronoPack revealed a striking, previously uncharacterized bifurcation in the NF- $\kappa$ B response emerging within minutes of stimulation (**Fig. 2a,e**). One subpopulation activates the classical inflammatory program, marked by strong induction of NF- $\kappa$ B hallmark genes like NFKB1A and ICAM1, whereas a smaller, erythroid-like subpopulation responds weakly (**Fig. 2c,d**). This rapid divergence indicates that intrinsic cell-state



heterogeneity preconfigures responsiveness to inflammatory signaling, influencing both the timing and magnitude of NF- $\kappa$ B activation.



**Figure 2.** (a) Overview of ChronoPack package workflow. ChronoPack is an integrated computational package for time-resolved single-cell analysis. It performs standard preprocessing, expression correction, and metacell construction, followed by trajectory inference using WaddingtonOT-derived transition kernels. The suite quantifies temporal gene trends with negative-binomial generalized additive models (NB-GAMs), reconstructs causal gene regulatory networks, and infers dynamic cell-cell communication through time-lagged ligand-receptor analysis. Together, these modules enable end-to-end analysis of ChronoSeq datasets from raw counts to causal dynamics. (b) UMAP of K562 scRNA-seq time series experiment based on derived transition kernel. (c) Volcano plot of highly varying genes depicting temporal trend scores (i.e.,  $\log_{10}(W_g)$ ) vs. maximum  $\log_2FC$ , where  $W_g$  is the Wald test statistic from tradeSeq's *associationTest*. Genes are labeled as significantly trending over time if  $q < 0.01$  at an FDR of 0.05 (calculated by applying Benjamini-Hochberg to the  $p$ -values associated with  $W_g$ ), and as upregulated or downregulated based on whether  $\text{abs}(\log_2FC)$  exceeds 0.5. Here,  $\text{abs}(\log_2FC)$  is taken to be the highest log-fold change observed across time points vs.  $t=0$ . (d) Gene trajectories for representative genes highlight the differential NF- $\kappa$ B response across K562 erythroid identity. (e) Metacell fate probabilities for the two response types based on the WaddingtonOT<sup>30</sup>-derived transition kernel. Terminal fates are identified as the last measured metacells in either response. (f) (left) Top 20 genes whose expression in early metacells correlated with fate propensity. (right) Expression vs. fate propensity plots for representative genes.

We next quantified gene-specific dynamics along each trajectory using negative-binomial generalized additive models (NB-GAMs). After filtering for genes with Time-Tag-specific capture biases, we identified 184 high-confidence temporally regulated genes, including both upregulated and a few immediate-early genes like ID3 and EGR1 which decreased in expression over time (Fig. 2c). The representative trajectories in Fig. 2d highlight the underlying biology: canonical NF- $\kappa$ B targets (e.g., NFKBIA, NFKB2, ICAM1, BIRC3, CXCL3) rise rapidly in the inflammatory trajectory but are dampened in the erythroid-like K562 cells, which express HBE1, HBA2, GYPC, and other lineage markers. These results indicate that pre-existing

erythroid identity constrains the amplitude and persistence of the NF- $\kappa$ B transcriptional response.

Finally, to identify genes associated with the differential NF- $\kappa$ B response, we designated the terminal metacells in each trajectory as distinct fates and calculated fate probabilities using the transition kernel. Since this kernel forms a directed acyclic graph, we computed fate probabilities by dynamic programming, first assigning probabilities to terminal cells and then recursively back-propagating these values to progressively earlier cells along the graph. Correlating gene expression with these probabilities revealed a set of early markers predictive of fate choice (**Fig. 2e-f**). Genes enriched in the inflammatory trajectory included immune and myeloid-like regulators such as KLF2A and CD3D, whereas the alternative trajectory was characterized by erythroid-associated genes like HBA1 and GYPA. These findings not only validate the observed phenotypic bifurcation but also identify potential regulators that may instruct this cell fate decision.

### **Minute-scale temporal data enables causal inference of gene regulation**

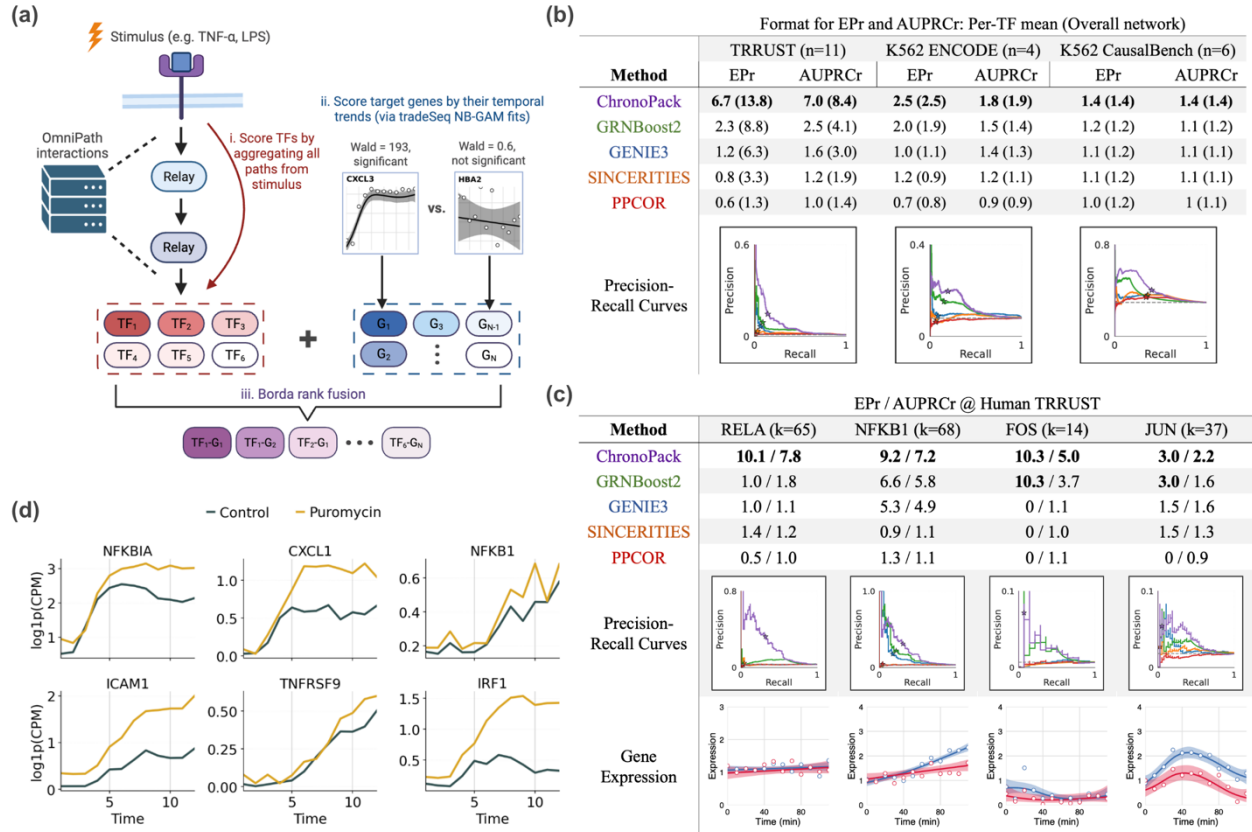
We next investigated whether ChronoSeq's dense temporal sampling could improve causal inference of gene regulatory networks (GRNs). The TNF- $\alpha$  stimulation of K562 cells kickstarts an intracellular signaling cascade that culminates in the rapid nuclear translocation of NF- $\kappa$ B (NFKB1-RELA) and the activation of AP-1 via MAPK signaling<sup>32</sup>. Traditional co-expression methods struggle to model transcription factors (TFs) with post-translational activity changes, such as RELA/p65 whose mRNA level remains stable during the NF- $\kappa$ B response. As part of the ChronoPack analysis suite, we have developed an inference strategy for application to rapid signaling contexts like NF- $\kappa$ B (**Fig. 3a**), in which the stimulus is conceptualized as an intervention impacting the activity of several unknown TFs. Our motivating hypothesis is that genes with significant temporal changes are enriched for direct targets of TFs whose activity shifts downstream of the stimulus. Following this logic, GRN inference proceeds by first (i) prioritizing TFs according to their downstream proximity to the stimulus, leveraging OmniPath<sup>33</sup> protein-protein interaction and pathway annotations to identify the TFs most likely to undergo activity changes, and (ii) ranking potential target genes according to the significance of their temporal expression dynamics. The resulting TF ranking captures both epistemic uncertainty about which TFs are perturbed as well as genuine biological multiplicity where several TFs may be affected by the same stimulus. TF and gene rankings are then combined additively to construct an edge list for network prediction, with a substantial rank penalty applied to edges terminating at genes without significant temporal trends (i.e.,  $q > 0.01$ ).

When benchmarked against four state-of-the-art algorithms (GENIE3, GRNBoost2, SINCERITIES, and PPCOR)<sup>34-37</sup>, ChronoPack achieved the highest early precision (EPr) and area under the precision-recall curve (AUPRCr) ratios across relevant TFs, including the NF- $\kappa$ B axis (e.g., NFKB1/RELA) and immediate-early AP-1 factors (FOS/JUN). Against curated human TRRUST interactions, ChronoPack demonstrates a nearly 3-fold improvement in mean performance over the best co-expression-based method, GRNBoost2 (**Fig. 3b**; EPr: 6.7 vs. 2.3, AUPRCr: 7.0 vs. 2.5). These predictions were independently supported by two K562-specific ChIP-seq networks derived from ENCODE<sup>38</sup> and taken from CausalBench<sup>39</sup>. Compared to TRRUST<sup>40</sup>, ChIP-seq-derived ground-truth networks, although data-driven and cell type-specific, are constrained by their strong context dependence and by the fact that physical TF-DNA binding does not necessarily equate to functional regulation. However, despite their lower specificity for true NF- $\kappa$ B regulation, we observed a consistent gain in EPr and AUPRCr on both ChIP-seq-based benchmarks. Interestingly, this improvement generalized to several TFs not expected to be differentially activated by TNF- $\alpha$  stimulation. This likely reflects the fact that the

regulons of these TFs are enriched for immediate early and stress-responsive genes. Erythroid-driving TFs like GATA1 and TAL1 also exhibited higher predictive performance, possibly due to overlap in their downstream targets with the NF- $\kappa$ B transcriptional program.

Per-TF performance makes the contrast between ChronoPack and co-expression baselines especially clear. For RELA (p65), ChronoPack outperforms other methods by a wide margin (**Fig. 3c**; AUPRCr: 7.8 vs. 1.8 @ GRNBoost2), because RELA's activity after TNF- $\alpha$  is driven by rapid post-translational control rather than a change in mRNA abundance – leaving co-expression methods with little signal to exploit. ChronoPack maintains high early precision ( $\sim 0.30$ ) by identifying perturbed TFs and prioritizing targets with clear temporal trends, whereas other methods are only marginally better than a random predictor. For NFKB1, the gap narrows (AUPRCr: 7.2 vs. 5.8 @ GRNBoost2) largely because NFKB1 is itself an NF- $\kappa$ B-inducible gene, resulting in apparent co-expression with true regulatory targets. Collectively, these findings demonstrate that high-temporal-resolution single-cell data from ChronoSeq improve the reconstruction of gene regulatory networks, revealing transcriptional control relationships that remain obscured in static datasets.

To distinguish direct TF-gene interactions from indirect responses, we repeated the TNF- $\alpha$  stimulation of K562 cells in the presence of puromycin, which blocks translation (**Fig. 3d**). Under this co-treatment, pre-existing NF- $\kappa$ B TFs can still translocate to the nucleus and bind DNA, but *de novo* synthesis of downstream transcriptional regulators and feedback inhibitors is largely prevented. Genes whose temporal trajectories are preserved in the presence of puromycin are therefore expected to be directly driven by stimulus-responsive TFs, whereas attenuated responses indicate dependence on newly translated intermediates. Consistent with this model, canonical NF- $\kappa$ B targets including NFKBIA, CXCL1, ICAM1, TNFRSF9, and IRF1, as well as NFKB1 itself, show robust, and in many cases augmented, induction with puromycin co-treatment – likely reflecting the loss of translationally-dependent negative feedback (e.g., reduced synthesis of I $\kappa$ B $\alpha$  and other inhibitors). These results provide functional support that ChronoPack-predicted regulatory edges, and more broadly, genes with significant temporal trends apparent within the first two hours of response, correspond predominantly to direct regulatory interactions.



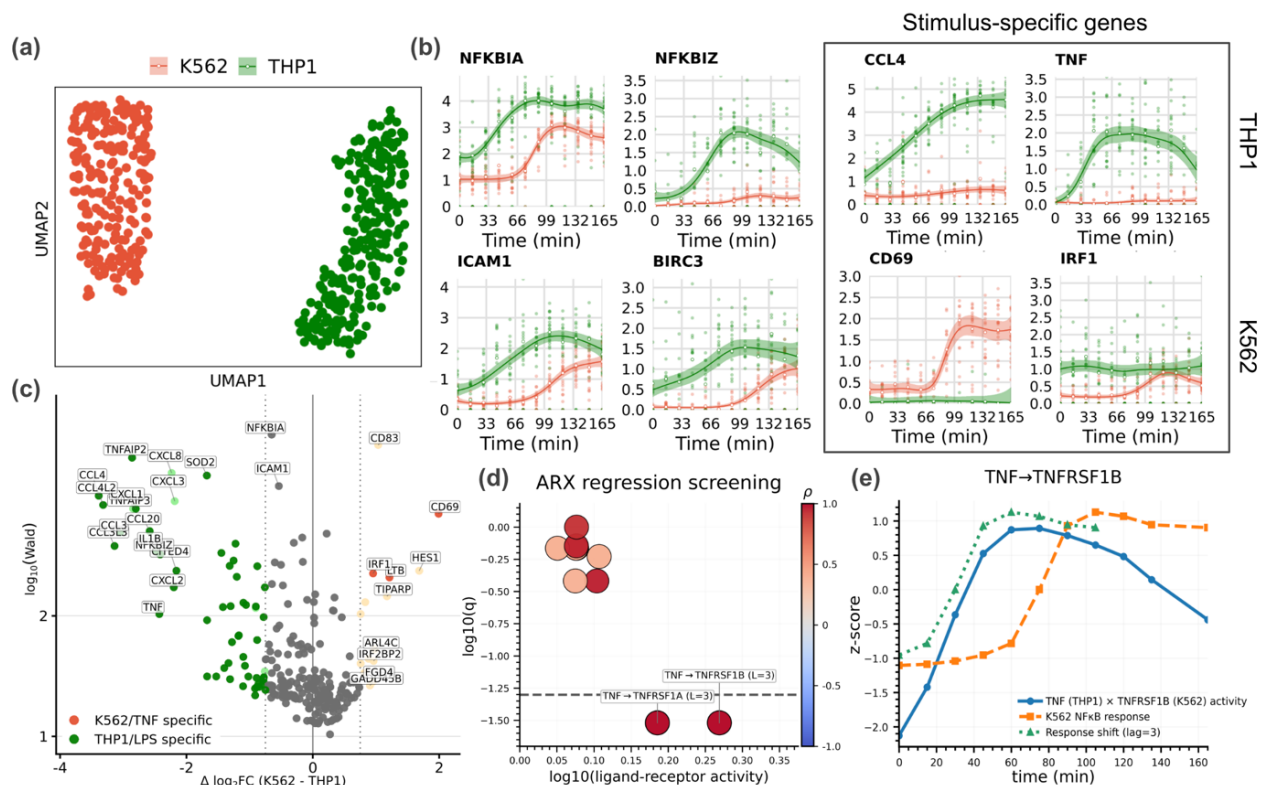
**Figure 3.** (a) Overview of ChronoPack GRN inference methodology. (b) Early precision (EPr) and area-under-precision-recall-curve ratios (AUPRCr) across inference methods and benchmark datasets (Human TRRUST<sup>40</sup> regulatory edges and two ChIP-seq-derived networks using either (i) K562 ChIP-seq experiments from ENCODE<sup>38</sup> or (ii) CausalBench's<sup>39</sup> K562 ground-truth).  $n$  = the number of NF- $\kappa$ B-relevant TFs evaluated and present in the ground-truth network. Reported metrics are formatted as per-TF mean (overall network), where the latter indicates performance for the entire subnetwork spanned by the NF- $\kappa$ B TFs. (c) Inference performance for individual NF- $\kappa$ B TFs.  $k$  = number of gene targets evaluated for the TF. (d) Mean expression curves for select NF- $\kappa$ B hallmark genes following TNF- $\alpha$  stimulation with and without puromycin addition. Lack of response attenuation with puromycin implies direct regulation.

## ChronoSeq recovers a paracrine TNF signaling cascade during multicellular immune activation

To test whether ChronoSeq can resolve dynamic cell-cell communication, we modeled innate immune secondary signaling via an interaction between LPS-responsive THP-1 macrophages and LPS-unresponsive K562 cells (**Fig. S7**). Following LPS stimulation, high-temporal-resolution profiling at 15-minute intervals revealed a sequential transcriptional response: THP-1 cells rapidly initiated a canonical inflammatory program, while K562 cells displayed a delayed, secondary response dependent on THP-1 co-culture (**Fig. 4a,b**). Two key features emerged: (i) a ~45 min lag in K562 activation relative to THP1, and (ii) the presence of cell type-specific gene programs. Differential temporal association and log<sub>2</sub> fold-change contrasts between THP1 and K562 (**Fig. 4c**) revealed an LPS/TLR4-biased module (e.g., *TNFAIP2*, *TNFAIP3*, *NFKBIZ*, *CCL3*, *CCL4*) versus a TNF-specific program (*CD69*, *IRF1*, *LTB*). These differences underscore the functional distinction between LPS-induced and paracrine TNF- $\alpha$ -induced NF- $\kappa$ B inflammatory responses in THP1 and K562, respectively. A subset of chemokines (*CXCL1*, *CXCL3*, *CXCL8*) were preferentially but not exclusively expressed in THP1 cells.

This experimental design allows us to prioritize ligand-receptor pairs which precede and are predictive of the K562 response. As seen from **Fig. 4b**, TNF expression rises early in THP1 and precedes the K562 NF- $\kappa$ B response by approximately 45 minutes (**Fig. 4e**). The temporal

ordering, specificity, and known TNF biology support a paracrine TNF relay from THP1 cells that drives the delayed K562 response<sup>41–43</sup>. To quantitatively identify candidate mediators of the delayed K562 activation, we used CellChatDB<sup>44</sup> annotations to compute ligand–receptor (L–R) activity scores (ligand in THP1, receptor in K562) over time and applied an autoregressive with exogenous input (ARX)/Granger causality<sup>45</sup> to screen L–R pairs against the K562 NF- $\kappa$ B response (**Fig. 4d**). Specifically, we tested whether the activity time series for a given L–R activity pair Granger-caused K562's NF- $\kappa$ B response across a plausible set of lags (between 30–75 min). This analysis prioritized pairs whose activity preceded and predicted K562 activation, where TNF–TNFRSF1A and TNF–TNFRSF1B emerged as top candidates. Consistent with this, TNF transcripts sharply increased in THP-1 cells within minutes of LPS exposure, preceding the late induction of NF- $\kappa$ B target genes such as ICAM1 and NFKBIA in K562 cells. These results demonstrate that ChronoSeq can temporally resolve and causally attribute multicellular signaling events, providing a direct systems-level view of how paracrine communication unfolds in real time.



**Figure 4.** Stimulation with  $1 \mu\text{g ml}^{-1}$  LPS occurred 5 min after the first sample, with subsequent samples collected every 15 min for 12 time points. At this LPS concentration, THP1 cells respond via TLR4 signaling<sup>46,47</sup>, whereas K562 cells remain unresponsive (**Fig S7**). (a) UMAP visualization of THP1 and K562 cells. (b) Gene expression trajectories for select genes demonstrating a lagged NF- $\kappa$ B response and stimulus-specific genes. (c) Volcano plot of significantly trending genes vs. their difference in maximum log2FC across the two cell lines. Darker colors are used to highlight genes with log2FC < 0.5 in the other cell/stimulus type, indicating specificity. (d) Granger-style regression screening of ligand-receptor candidates. (e) Visualization of TNF-TNFRSF1B ligand-receptor activity vs. K562 NF- $\kappa$ B response

## Discussion

We have demonstrated that ChronoSeq addresses a practical bottleneck in time-resolved single-cell transcriptomics. By automating sample collection across time points and consolidating them into a single library via time-barcoded beads, ChronoSeq achieves minute-level sampling of expression dynamics over multiple hours. Automation and the immediate cell

lysis help eliminate human errors in sampling and labelling, while integrated cell culture helps maintain optimal cell viability and function for experiments. Time-tag separation is clear in bulk and single-cell human-mouse mixes with little cross contamination; doublets remain low under standard loading; and transcript measurements are concordant with 10x Genomics Chromium and bulk references. ChronoSeq also shows high similarity to qPCR measurements of K562 cells stimulated with TNF- $\alpha$ , providing validation with an orthogonal approach. Collectively, these data show that ChronoSeq can introduce reliable time labels at unprecedented time resolutions while maintaining accuracy, cell viability, high throughput and dramatically cutting down on labor requirements for time labelled scRNA-seq.

Applied to TNF- $\alpha$ -stimulated K562 cells, minute-scale sampling resolves an early bifurcation into an inflammatory trajectory with robust NF- $\kappa$ B target induction and a weaker, erythroid-like trajectory. By fitting NB-GAMs within each trajectory, we were able to identify genes with significant temporal dynamics in a principled way. The erythroid branch exhibited distinct kinetics – e.g., transient pulses in CD69 and BIRC3, whereas in the inflammatory branch these genes plateau or continue to rise. For regulatory inference, ChronoPack outperforms traditional methods against both TRRUST and K562 ChIP-seq-based reference networks. This advantage is most pronounced for TFs that are regulated post-translationally without obvious changes in mRNA expression, such as RELA.

In a mixture experiment of K562 and THP1 cells stimulated by LPS, ChronoSeq measures expression profiles that enable the inference of cell-cell signaling. We observed a ~45-minute delay in the K562 NF- $\kappa$ B response compared to THP1, originating from a difference in TLR4 expression by these two cell types. Combining time-resolved ligand-receptor activity scores with a simple ARX/Granger screen nominates mediators that both rise early in THP1 and predict K562 dynamics. Here, we see that TNF interactions with both TNFRSF1A and TNFRSF1B rank highly, recovering established TNF paracrine biology. Importantly, at longer sampling intervals, the relationship between ligand expression and NF- $\kappa$ B response would have been far less evident, limiting the inference of cell signaling from observable lags. This workflow shows how ChronoSeq can prioritize mechanistic hypotheses of cell-cell communication from temporal single-cell transcriptomic measurements alone.

In the context of recent efforts to predict cellular responses to various interventions (drugs, cytokines, and genetic perturbations; see Arc Institute's Virtual Cell initiative<sup>2</sup>), ChronoSeq provides a complementary, high-resolution view of the immediate transcriptional dynamics following a stimulus. In turn, this allows us to better distinguish direct regulation from indirect effects – an ambiguity that persists when GRNs are inferred from single-snapshot assays such as Perturb-seq.

While ChronoSeq represents a significant advance, its true power will be realized in its application to more complex and physiologically relevant systems. The current throughput, while sufficient for cell lines and major immune populations, may present challenges for profiling very rare cell types in heterogeneous tissues. Furthermore, the transcriptome is only one layer of regulation; a complete "Virtual Cell" model will require future integration with technologies capturing protein activity, chromatin state, and metabolic fluxes at comparable resolution. The logical next step is the fusion of ChronoSeq with precise perturbations to dynamically map the minute-scale consequences of genetic or chemical interventions and build fully predictive models of cellular behavior.



## Data Availability

ChronoSeq data are available in the ArrayExpress database (<http://www.ebi.ac.uk/arrayexpress>) under accession numbers E-MTAB-15894, E-MTAB-15928, E-MTAB-15927, E-MTAB-15946, E-MTAB-15947, E-MTAB-15956, E-MTAB-15955, E-MTAB-15964, E-MTAB-15965 and E-MTAB-15966. Parameters used for each ChronoSeq experiment and the associated accession numbers are listed in **Table 1**.

| Table 1: Experimental parameters used for various ChronoSeq datasets and their accession numbers |                    |           |                                                |                   |                                                  |                                        |
|--------------------------------------------------------------------------------------------------|--------------------|-----------|------------------------------------------------|-------------------|--------------------------------------------------|----------------------------------------|
| Experimental condition                                                                           | Cell concentration | Time-Tags | Stimulus                                       | Sampling interval | Stimulation time                                 | EMBL-EBI ArrayExpress accession number |
| K562                                                                                             | 430 cells/ $\mu$ l | 12        | 10ng/ml TNF- $\alpha$                          | 10 minutes        | 5 minutes after first sample                     | E-MTAB-15894                           |
| 50:50 K562 + differentiated THP1 cells                                                           | 430 cells/ $\mu$ l | 12        | 1 $\mu$ g/ml LPS                               | 15 minutes        | 5 minutes after first sample                     | E-MTAB-15928                           |
| K562                                                                                             | 500 cells/ $\mu$ l | 12        | 10ng/ml TNF- $\alpha$ + 5 $\mu$ g/ml Puromycin | 10 minutes        | Both added together 5 minutes after first sample | E-MTAB-15947                           |
| K562 Negative Control (Reverse Time-Tag injection order 12 to 1) (QC)                            | 860 cells/ $\mu$ l | 12        | No Stimulus. Time-Tag Bias estimation          | 10 minutes        | N/A                                              | E-MTAB-15946                           |
| 50:50 K562+EL4 (QC)                                                                              | 215 cells/ $\mu$ l | 3         | 10ng/ml TNF- $\alpha$                          | 10 minutes        | 5 minutes after first sample                     | E-MTAB-15955                           |
| 50:50 K562+EL4 (QC)                                                                              | 430 cells/ $\mu$ l | 3         | 10ng/ml TNF- $\alpha$                          | 10 minutes        | 5 minutes after first sample                     | E-MTAB-15956                           |
| 50:50 U937+K562                                                                                  | 430 cells/ $\mu$ l | 12        | 0.5 $\mu$ g/ml LPS                             | 10 minutes        | 5 minutes after first sample                     | E-MTAB-15927                           |
| K562+EL4 (Time Tag 10), K562(Time Tag 11), EL4 (Time Tag 12) (QC)                                | 215 cells/ $\mu$ l | 3         | No Stimulus. Only Checking Time-Tags           | 7 minutes         | N/A                                              | E-MTAB-15964                           |
| Bulk K562 and EL4 cells mixed with 12 Time-Tags for verification (QC)                            | 600 cells/ $\mu$ l | 12        | No Stimulus. Only Checking Time-Tags           | N/A               | N/A                                              | E-MTAB-15965                           |
| Bulk K562 cells negative control (QC)                                                            | 600 cells/ $\mu$ l | 12        | No Stimulus                                    | 10 minutes        | N/A                                              | E-MTAB-15966                           |

## Online Methods

### Experimental procedures:

Detailed experimental procedures and protocols can be found on the ChronoSeq project website <http://wanglab.ucsd.edu/ChronoSeq/>. The notebooks associated with the various protocols also feature embedded YouTube videos. See **Supplementary Table S2** for items purchased for assembling the ChronoSeq device. For the latest and most accurate links and information always refer to the protocols and instructions on the project website.

### Device design overview

Our device design (**Figure S8**) addresses several challenges not present in the original Drop-seq<sup>7</sup> system. See **Figure S9** for a comparison and detailed discussion of the ChronoSeq and Drop-seq devices. The system distinguishes itself from conventional Drop-seq devices through several key innovations: automated bead handling with vortex suspension, integrated cell

culture system with 37°C water bath and 5% CO<sub>2</sub> environment, robotic droplet collection and bead recovery system, and compressed air pressure control with flow sensing for rapid flow rate changes.

### Microfluidic chip design and fabrication

See **Figure S10** for a comparison of the ChronoSeq and Drop-seq chip designs and the reasons for the various design changes. For fabrication standard photolithography and PDMS molding techniques<sup>48</sup> are used for chip fabrication. Chips are plasma bonded to glass slides and chemically treated to create hydrophobic channels. We implemented a rigorous quality control protocol: compressed air testing to check for delamination and eliminate leaking chips, simulating the high pressures used during the flushing phase; visual inspection to ensure the absence of particles before connecting to the device. These design and quality control measures enable the chips to withstand higher pressures and cyclical pressurization during the flushing phase. The optimized chip design is crucial for reliable, high-throughput, time-resolved single-cell sequencing. The improvements in channel geometry and manufacturing process contribute to reduced clogging, increased cell capture efficiency, and overall system robustness. These features are essential for maintaining consistent performance across multiple time points and ensuring high-quality data collection throughout extended experimental runs.

### Creation of Time-Tagged ChronoSeq beads from Drop-seq beads

ChronoSeq beads were prepared by modifying Drop-seq beads (**Supplementary Figure S3**). The modification principle involves binding oligonucleotides (SEQ1 through SEQ12 **Supplementary Table S3**) to the existing polyT region of Drop-seq beads, followed by extension with *E. coli* DNA Polymerase I to create the reverse complement of the oligo, and subsequent alkaline denaturation to generate single-stranded DNA. Exact protocol used can be found in **Supplementary Protocol 1**.

### Library preparation

ChronoSeq uses the same oligos used for Drop-seq library preparation (**Supplementary Table S3**). ChronoSeq library preparation protocol (**Supplementary Protocol 2**) has an expected yield between 1.5-6ng/  $\mu$ L after the first PCR reaction as measured by Qubit. While Tagmented sequencing-ready libraries have an expected concentrations of 20-50 ng/ $\mu$ L. Libraries were sequenced on one lane of Illumina NovaSeq X Plus 10B with 10% PhiX Spike-In and Custom Read 1 Primer mixed with Standard Illumina Primers. Either PE100 or PE150 configurations were used for the sequencing run. The ChronoSeq protocol is very similar to the Drop-Seq V3.1 protocol with the following notable changes. First, the Exonuclease 1 treatment step was omitted since it did not contribute to Library yield, quality or capture rates. Next, 14 PCR cycles were typically done instead of 12 PCR Cycles recommended by Drop-Seq for the first round of PCR. Finally, beads were split into different reactions based on number of Timepoints instead of number of estimated STAMPs for both PCR and Reverse Transcription.

### Cell culture

All cell lines were cultured in DMEM supplemented with 10% fetal bovine serum (FBS) and 1% penicillin-streptomycin (Pen-Strep) at 37°C in 5% CO<sub>2</sub>. THP-1 cells were additionally differentiated into macrophages with 100 nM phorbol 12-myristate 13-acetate (PMA) for 48 hours, then rested for 72 hours. THP-1 macrophages were then converted to a cell suspension



non-enzymatically with EDTA to preserve cell surface proteins. **Supplementary Protocol 3** provides exact details of cell culture conditions for each cell line.

### qPCR validation

K562 cells were sampled with the ChronoSeq device at 10-minute intervals over 2 hours, yielding 12 timepoints. Following baseline sampling, TNF- $\alpha$  (10 ng/ $\mu$ L) was introduced at the 5-minute mark to induce inflammatory signaling. Cell lysates were prepared using a modified Bio-rad SingleShot Cell Lysis Kit protocol (**Supplementary Protocol 4**) in which the cell suspension was directly mixed with the Lysis reagents instead of washing and changing the media to DPBS.

### qPCR validation of K562 co-stimulation by THP1 cells and detection of RNAase activity

To investigate paracrine signaling effects between immune cell populations, THP-1 cells were differentiated and stimulated with 1  $\mu$ g/ml LPS, with cell supernatants collected at 0, 1, 2, and 3 hours post-stimulation. These time-course supernatants were then applied to K562 target cells. Controls included unstimulated K562 cells and K562 cells stimulated with TNF- $\alpha$ . K562 cells were sampled at 0 and 1 hour following supernatant addition using a modified Bio-rad SingleShot Cell Lysis Kit protocol (**Supplementary Protocol 5**). Next, we performed qPCR for the relatively lowly expressed housekeeping gene HPRT1 in some of the same samples of K562 cells stimulated with THP1 supernatants and controls (**Fig. S7**). HPRT1 was not detected in all samples stimulated with THP1 supernatants, with no Cq values generated (See **Table S1**). Based on the same data we also observed no RNAase activity for K562 cells stimulated with TNF- $\alpha$ . This confirmed high RNAase activity is a problem unique to THP1 cell suspensions. Thus, we added 3U/ $\mu$ l of Watchmaker Genomics RNAase inhibitor at 800U/ $\mu$ l was added to each Time-Tag for our THP1-K562 mixture experiment.

## Computational procedures:

### ChronoSeq-Tools pipeline

The ChronoSeq-Tools pipeline extends the Drop-seq pipeline methodologies to enable temporal tracking of single-cell transcriptomes through integrated Time-Tags. The pipeline processes paired-end RNA sequencing data through sequential barcode extraction, alignment, and Time-Tag determination phases to generate a final Digital Gene Expression matrix (DGE) with Time-Tags for each Cell. Scripts and instructions for running the ChronoSeq-Tools pipeline can be found on GitHub <https://github.com/kanishkasthana/ChronoSeq-Tools>. Moreover, code for generating QC plots in this paper can be found at <https://github.com/kanishkasthana/ChronoSeq-QC>

### Pipeline architecture

The workflow initiates by converting FASTQ files to unaligned BAM format using Picard's FastqToSam. Three sequential tagging steps employ Drop-seq's TagBamWithReadSequenceExtended to isolate:

1. **Cell barcodes** (XC tag: bases 1-12)
2. **Unique molecular identifiers** (XM tag: bases 13-26)

3. **Time-Tags** (YT tag: bases 27-86) with relaxed quality thresholds ( $\leq 150$  low-quality bases permitted).

Quality control steps include filtering reads with low-quality tags and trimming adapter sequences (AAGCAGTGGTATCAACGCAGAGTGAATGGG) and polyA tails using Drop-seq tools. Processed reads undergo STAR alignment against species-specific references, followed by queryname sorting and MergeBamAlignment to preserve metadata. Gene annotation occurs via TagReadWithGeneFunction using RefFlat files, generating digital expression matrices (DGEs). Computational efficiency is enhanced through optional scratch directory usage and distributed locking mechanisms to manage I/O operations on SLURM cluster systems.

#### Alignment references

For species-mixing experiments reads were aligned to the same hg19-mm10 mixed cell reference used in Dropseq. For all other experiments hg38 was used for alignment.

#### Time-Tag determination

Time tags derive from twelve synthesized oligonucleotides containing a combination of degenerate bases (R,Y,M,K,S,W,H,B,V,D,N) and constant bases flanked by polyA sequences. The GetTimeTags algorithm:

1. Extracts YT tag sequences from aligned BAM files
2. Performs reverse complement conversion
3. Matches sequences against regex patterns for each Time-Tag.
4. Requires  $\geq 20$  detections per cell barcode for reliable assignment.
5. Assigns final Time-Tags through majority vote ( $\geq 70\%$  prevalence) for each Cell barcode. Cell barcodes failing to meet this threshold are flagged as having "Time Tag Collision Detected" indicating potential cross-contamination or sequencing errors

#### Comparison with Drop-seq pipeline

While sharing core components (Picard, Drop-seq tools, STAR alignment), ChronoSeq introduces three key modifications:

| Feature               | Drop-Seq            | ChronoSeq                             |
|-----------------------|---------------------|---------------------------------------|
| Barcode Structure     | 12bp cell + 8bp UMI | 12bp cell + 14bp UMI + 60bp time tag  |
| Tag Quality Threshold | Strict (1-2 errors) | Permissive for time tags (150 errors) |
| Output                | Gene expression     | Expression + temporal stratification  |

#### Optional error correction

ChronoSeq supports optional error correction for both substitution and synthesis errors in cell barcodes, closely mirroring Drop-seq's established methodology. Our substitution error correction has been optimized to run on multiple cores to speed up execution. Like Drop-seq the correction step identifies cell barcodes differing by a single base (Hamming distance of 1) and

collapses those with a dominant sequence representing at least 93% of reads, mitigating sequencing errors. The synthesis error correction step identifies and collapses barcodes with systematic errors at the barcode termini, typically arising from bead synthesis imperfections. This is accomplished by constructing a Prefix-Trie data structure and merging barcodes that differ only at the last few positions, replacing the error-prone bases with 'N' and updating associated UMIs accordingly. Using a Prefix-Trie approach allows us to speed up execution from Quadratic to Linear time for our approach compared to Drop-Seq. These corrections are performed prior to digital gene expression quantification and time tag assignment, ensuring high-fidelity cell barcode recovery

### ChronoPack package:

**Standard preprocessing.** Time-Tagged ChronoSeq DGE data generated using the ChronoSeq-Tools pipeline was processed using the standard Scanpy<sup>49</sup> workflow. Quality filters were applied to remove cells with too few counts (<200 genes, <500 UMIs), as well as potential doublets (>15000 UMIs) or damaged cells with mitochondrial expression percentage >7.5. Cells predicted to be doublets via Scrublet<sup>50</sup> were also removed. The filtered gene-count matrix was library normalized and log transformed (sc.pp.normalize total, sc.pp.log1p). Each cell was assigned S and G2M phase scores using sc.tl.score genes cell cycle based on cell cycle marker genes from Kowalczyk et al, 2015<sup>51</sup>. The dataset was subset to the top 3000 highly varying genes using sc.pp.highly variable genes.

**Expression correction.** Technical factors (total UMIs, percentage of mitochondrial expression) and cell cycle effects (S and G2M scores) were regressed out prior to applying ComBat<sup>52</sup> to correct for different reverse transcription batches. The corrected expression matrix was scaled using sc.pp.scale, followed by sc.tl.pca to calculate principal components (PCs). Reduced embeddings for each cell were selected as the top n PCs, where the optimal value of n was selected by plotting PC variance ratios and identifying the elbow (i.e., the point of maximum curvature).

**Metacell construction.** Similarly to Sheu et al, 2024<sup>53</sup>, metacells were constructed via k-means clustering within each time point using these embeddings. A default value of  $k = (20 \times \text{number of cell types})$  was used. Raw counts for cells assigned to the same metacell were aggregated, followed by library normalization and log transformation. Importantly, no obvious batch effects were observed in the resulting metacells as they were aggregated from cells that were similar in the corrected expression space.

**Deriving a transition kernel from optimal transport couplings.** WaddingtonOT<sup>30</sup> was used to infer transport couplings between metacells across successive time points (i.e. from  $t$  to  $t+1$ ). Default parameters for the entropy, row sum, and column sum regularization terms were used ( $\epsilon=0.05$ ,  $\lambda_1=1$ ,  $\lambda_2=50$ ). The resultant couplings were stitched together into a sparse transport matrix  $T$  with  $T_{ij}$  encoding the descendant mass sent from cell  $i$  to cell  $j$ . The forward transition kernel is the row-normalized form of this matrix, calculated as  $P = D^{-1}T$  where  $D = \text{diag}(T1)$ , and  $P_{ij}$  is the probability of cell  $i$  transitioning to cell  $j$ .  $P$  enables us to (a) map cell fates and (b) impute gene expression changes from forward and backward expectations on  $P$ .

**Cell-wise trajectory clustering.** We hypothesized that the transition kernel could be used to cluster cells by their temporal responses. The expected fate distribution from a random walk of length  $k$  can be calculated as  $S_k = P^k$  where  $S_{ijk}$  represents the probability that cell  $i$  walks to cell  $j$ . A weighted average across length  $k = 1 \dots K$  random walks can then be constructed as  $S = \text{sum}$

$\lambda^k S_k$ . Here,  $\lambda$  is a hyperparameter that can be tuned to discount longer random walks; smaller values of  $\lambda$  favor higher  $S_{ij}$  for cells that are temporally proximal. Finally, this matrix was symmetrized as  $W = (S + S^T)/2$  such that higher edge weights corresponded to metacell pairs  $i, j$  if either one was reachable from the other across multiple steps forward in time. Finally, the Leiden algorithm was applied to  $W$  to assign clusters corresponding to different trajectories. These clusters on  $W$  correspond to communities of metacells that share similar futures and pasts according to  $P$ .

**Cell fate mapping.** Terminal fates were identified as the last metacells belonging to either trajectory cluster. Fate probabilities are traditionally calculated by converting  $P$  to an absorbing Markov chain and solving the linear system  $(I - Q)X = R$  where  $Q \equiv$  transitions between transient states,  $R \equiv$  transitions from transient to absorbing states (i.e., terminal fates), and  $I$  is the identity matrix<sup>54</sup>. However, since WaddingtonOT edges are strictly directed forward in time,  $P$  forms a directed acyclic graph for which fate probabilities can be estimated through a dynamic programming approach. Let  $p_k(i)$  denote the probability that a random walk starting at cell  $i$  is ultimately absorbed in terminal fate  $A_k$ . By setting the boundary conditions  $p_k(i) = 1$  for  $i \in A_k$  and  $p_k(i) = 0$  for  $i \notin A_k$ , a reverse-time sweep can be performed recursively via  $p_k(i) = \sum_j P_{ij} \cdot p_k(j)$  to recover fate probabilities  $p_k$  for all cells. Fate uncertainty was then calculated as the Shannon entropy  $H(i) = -\sum_k p_k(i) \cdot \log(p_k(i))$ . Finally, we identified early genes that were predictive of trajectory branch choice by correlating per-metacell expression in the earliest time bin with a fate-contrast score. Specifically, for each early metacell  $i$ , we defined the fate contrast as  $y_i = p_1(i) - p_0(i)$ . After removing genes detected in  $<10\%$  of early cells, Spearman rank correlations were computed. P-values were adjusted for multiple testing via Benjamini-Hochberg. Significant positively and negatively correlated genes were labeled as being associated with fates  $A_1$  and  $A_0$ , respectively.

**Identifying genes with significant temporal trends.** For each gene, the tradeSeq<sup>31</sup> package was used to fit a negative-binomial generalized additive model (NB-GAM), where mean expression is modeled as a cubic smoothing spline function over real time. Metacell counts are modeled with a negative binomial distribution  $Y_{gi} \sim \text{NB}(\mu_{gi}, \phi_g)$ , where  $\log(\mu_{gi}) = \sum_l s_{gl} Z_{li} + \log(N_i)$ . Here,  $N_i$  are library size offsets,  $Z_{li}$  denotes metacell membership in each lineage, and  $s_{gl} = \sum_k b_k(t) \beta_{g|k}$  is a smoothing spline constructed from cubic basis functions  $b_k(t)$ . A basis size of  $K=8$  was used. After fitting, tradeSeq's associationTest function was used to compute a Wald test statistic assessing the null hypothesis that all coefficients are equal, i.e.,  $H_0: \beta_{g|k} = \beta_{g|l}$ . The associated p-values were adjusted for multiple testing via Benjamini-Hochberg, and genes with  $q < 0.05$  were labeled as significantly trending over time. These genes were clustered by the similarity of their trajectory-wide expression patterns using tradeSeq's clusterExpressionPatterns method.

**Gene regulatory network inference.** Traditionally, GRN inference methods predict a ranking of TF-gene edges which can then be benchmarked against a ground-truth network. In our experiments, stimuli such as TNF- $\alpha$  or LPS induce changes in gene expression primarily through the post-translational modification of TF activity. In this context, co-expression modeling cannot be used to infer regulatory edges, as is done in GRNBoost2 and similar methods. Instead, we developed an inference strategy that independently ranks (i) TFs as being downstream of a stimulus (i.e., which TFs are most likely to be perturbed) and (ii) target genes as being impacted by that same stimulus, prior to fusing these rankings together via the Borda method to infer TF-gene regulatory edges.

To score TFs as being perturbed by a stimulus, we utilized PPI and signaling pathway annotations from OmniPath<sup>33</sup>. We first construct a directed, signed intracellular signaling

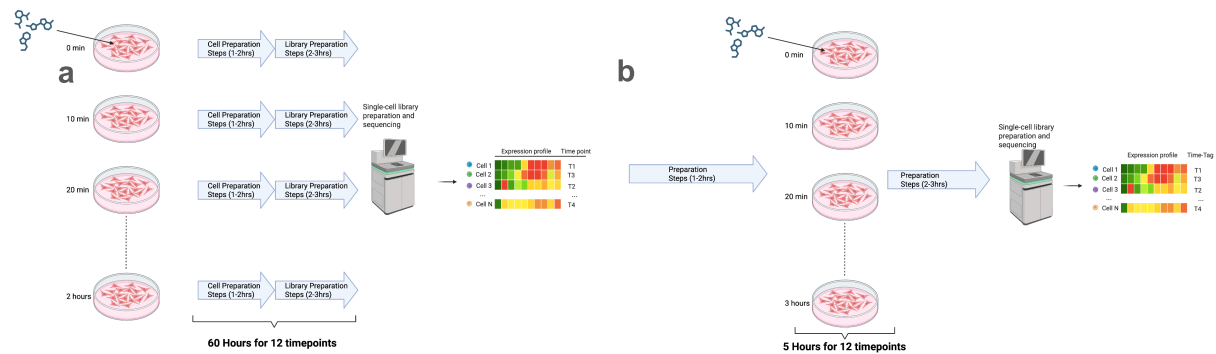
network centered on the stimulus of interest (e.g. TNF) using OmniPath's PostTranslational interactions. Edge signs in this directed network were encoded as +1 (stimulation), -1 (inhibition), or 0 (unknown) based on OmniPath's *is\_stimulation/is\_inhibition* flags. An expression filter was applied to restrict the graph to proteins that were likely to be active, where nodes were excluded if their corresponding RNA transcripts were detected in fewer than 1% of cells. Next, we enumerated all directed signaling paths from the ligand to each TF and summarized these paths into TF-level scores. We performed a depth-first traversal starting at the ligand (i.e., stimulus) of interest: the first hop used curated ligand-receptor pairs from CellChatDB, and all subsequent hops followed the directed, signed OmniPath graph up to a fixed radius of 5 steps. TF nodes were treated as sinks, so paths terminated once they reached a TF. For each distinct stimulus  $\rightarrow$  TF path, we tracked the cumulative sign  $s_i$  (i.e., the product of edge signs) from the stimulus to the TF. Each path was assigned a weight that decayed exponentially with path length, allowing for shorter paths to contribute more, and was also multiplied by a branching penalty  $B$  that downweighted routes passing through higher-degree nodes ( $W_i = B_i \delta^{L_i-1}$  where  $\delta = 0.7$ ). The final score for the TF was taken as  $\sum s_i W_i$ , summing the weights and cumulative signs of all paths  $i$  terminating at that TF. Target genes were ranked by the Wald statistic corresponding to tradeSeq's *associationTest*, where larger values corresponded to a more significant temporal trend. The TF- and target gene-specific rankings were then combined via a modified version of the Borda method to produce an overall ranked TF-gene edge list.

GRN inference was benchmarked against other co-expression-based methods, including GRNBOOST2, GENIE3, SINCERITIES, and PPCOR. Early precision ratio (EPr) and area-under-precision-recall-curve ratio (AUPRCr) were calculated to assess the performance of the predicted networks. We focused on evaluating these metrics per TF subnetwork, as well as the subnetwork spanned by all NF-kB-relevant TFs, provided that these TFs were also available in the ground-truth network. We considered three ground-truth networks: (a) the TRRUST<sup>40</sup> database, which contains literature-curated human TF-gene interactions, and (b-c) two K562-specific networks derived from ChIP-seq experiments. For the latter category, ChIP-seq peaks were either (b) retrieved from the ENCODE<sup>38</sup> project for the human K562 cell line, corresponding to 1670 experiments and 757 unique accessions or (b) taken directly from CausalBench<sup>39</sup>, which constructed its own ground-truth network in which edges were further filtered according to perturbation effect sizes.

**Cell-cell communication.** Ligand-receptor (L-R) pairs annotated as 'Secreted Signaling' were queried from CellChatDB<sup>44</sup>. In our experiment, the THP1 NF-kB response preceded that of the K562 cells, so we focused specifically on pairs where the ligand and receptor were adequately expressed (>5% with nonzero counts) in the THP1 and K562, respectively. We also required the corresponding ligands to vary significantly over time per our earlier NB-GAM statistics ( $q < 0.05$ ). This filtering procedure resulted in 10 candidate ligand-receptor pairs out of an initial set of 1199 interactions. Next, an L-R activity series  $A_{LR}(t) = \text{sqrt}(\mu^{\text{THP1}}_L(t) \mu^{\text{K562}}_R(t))$  was constructed as the geometric mean of the fitted means on the log1p scale. The K562 NF-kB response series  $y(t)$  was calculated as the per-time-bin average of a curated NF-kB module. To assess the relationship between the L-R activity and NF-kB response in K562 cells, we conducted a single-lag Granger<sup>45</sup> screening. For each lag  $L \in [2, 5]$  and candidate L-R pair, we fit an ARX (autoregressive with exogenous input) model  $y_t = c + (\alpha \cdot y_{t-1}) + (\beta \cdot x_{t-L}) + \epsilon_t$ . We tested  $H_0: \beta = 0$  via a partial F-test, comparing the restricted model (i.e., no  $x_{t-L}$ ) against the unrestricted model. To limit the space of feasible lags, we enforced a sign constraint by discarding lags with a negative cross-correlation  $\text{cor}(y_t, x_{t-L})$ , corresponding to the assumption that the presence of a ligand, rather than its absence, induces a signaling response in the K562 cells. The Holm-adjusted within-pair p-values were subjected to Benjamini-Hochberg (BH) correction across

pairs. While pairs with  $q < 0.05$  were considered statistically significant for visualization purposes, we primarily focused on the utility of these test statistics for prioritizing L-R pairs as candidates for cell-cell communication. Alternatively, we performed a lead-lag correlation test to assess the association between  $A_{LR}$  and  $y$ , which produced a similar ranking of the L-R pairs.

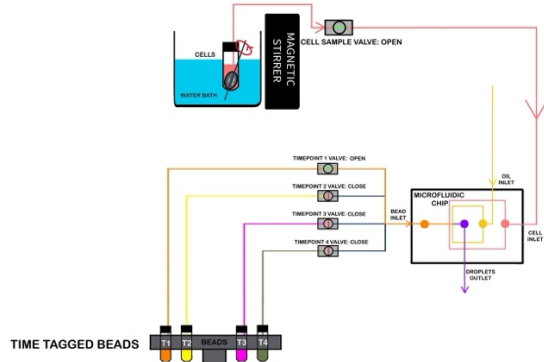
## Supplementary Figures and Tables



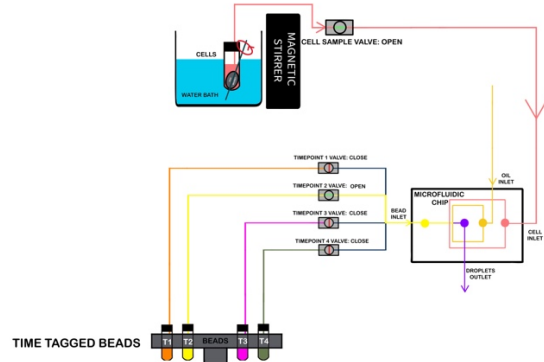
**Supplementary Figure S1.** a) Shows the current paradigm for a temporal drug response study. If twelve timepoints are taken over two hours, each timepoint would have to be processed independently. Optimistically if five hours of hands-on labor is needed for each timepoint, this would require a minimum of 60 hours of total hands-on labor for all timepoints. Due to the lack of automation and manual nature of the technology there is a high possibility of inconsistent sampling or labelling times for each time point. For example, in each sample the metabolites used for labelling need to be washed out. Washing out is not instantaneous and the cells have time to incorporate more metabolites during processing delays; thereby increasing the chance of introducing an error of several minutes or longer for each sample. An error of several minutes introduces the high possibility of measuring spurious perturbations, especially if the sampling rate is also several minutes. Therefore, the high labor requirement, lack of automation, separate labeling of each sample, no immediate cell lysis to preserve transcriptome and resulting minute-scale sampling/labelling errors, make it practically impossible to use metabolic labelling for profiling real-time perturbations over several hours. b) If this same temporal drug response study is done using ChronoSeq this would require only five hours of hands-on labor. Moreover, by automating sample collection across time points and consolidating them into a single library via time-barcoded beads, ChronoSeq achieves minute-level sampling of expression dynamics over multiple hours. Automation and the immediate cell lysis help eliminate human errors in sampling and labelling, while integrated cell culture helps maintain optimal cell viability and function for experiments.



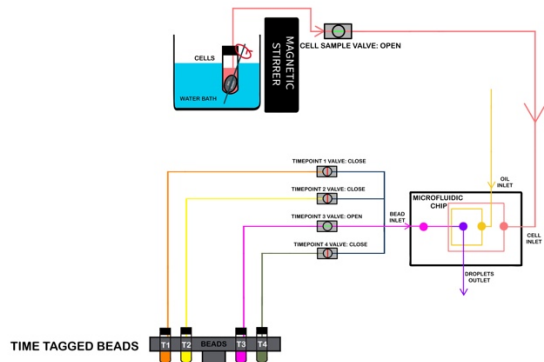
### First Timepoint Injection



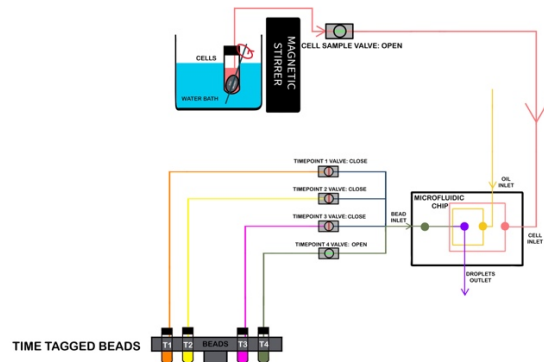
### Second Timepoint Injection



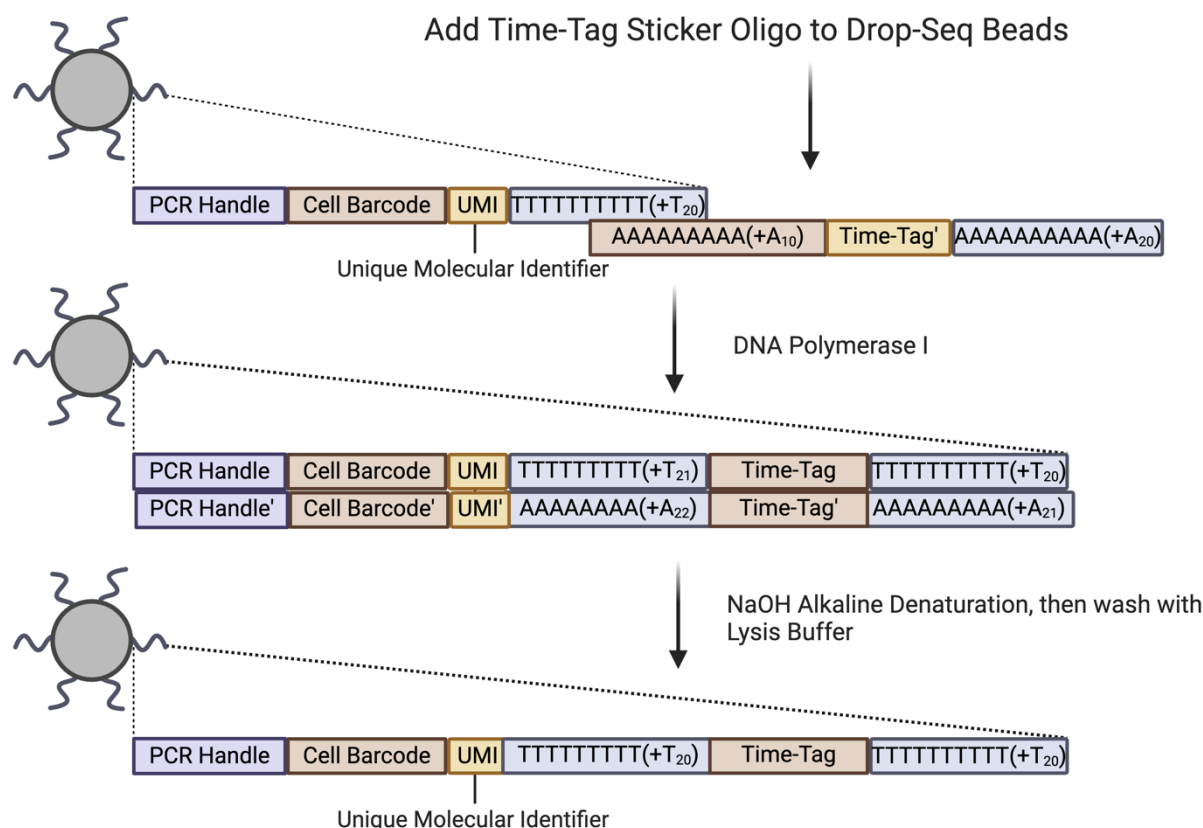
### Third Timepoint Injection



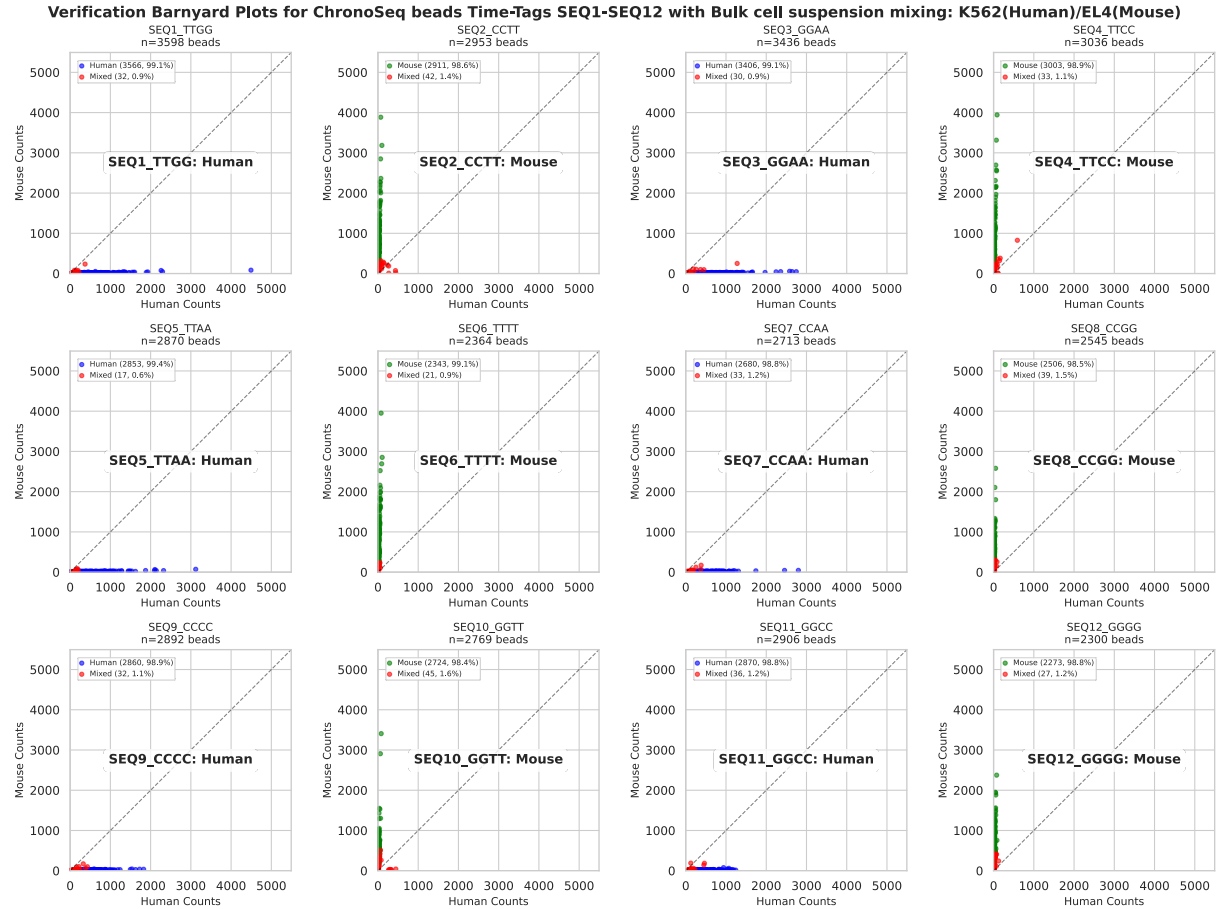
### Fourth Timepoint Injection



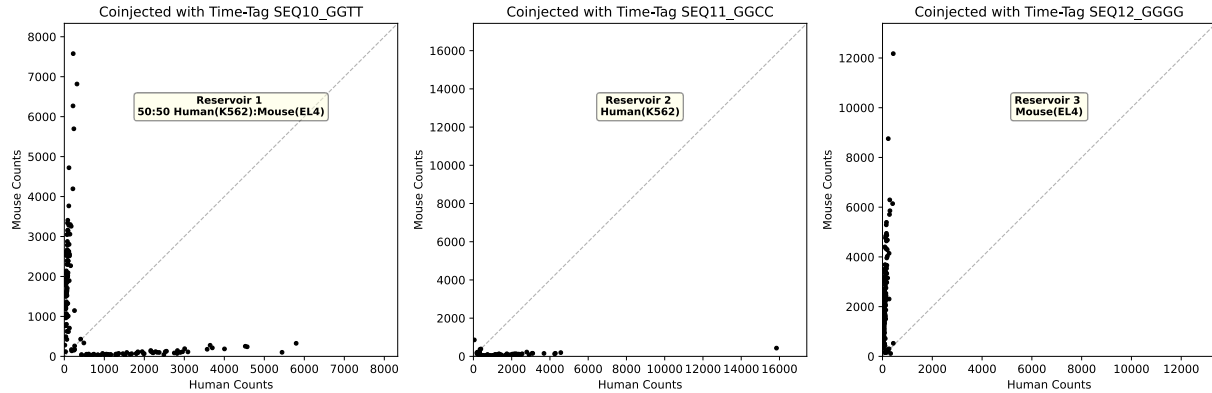
**Supplementary Figure S2: Simplified schematics to explain temporal barcoding using ChronoSeq device.** The device sequentially co-injects Time-Tagged beads and cells into a droplet-generating microfluidic chip at regular time intervals. These beads are stored in separate reservoirs (e.g., 4 reservoirs for 4 time points for illustration purpose), with each reservoir containing beads with a unique Time-Tag. For each timepoint, the Cell Sample Valve remains open while the corresponding Timepoint Valve (orange for T1, yellow for T2, Hot Pink for T3, and Olive Green for T4) is opened. This allows for the co-injection of time-specific beads with cells, resulting in libraries barcoded with distinct Time-Tags. This process enables precise temporal labeling of cells across different timepoints. Our system currently supports 12 unique Time-Tags with minimum 7-minute intervals including a 1-minute sampling duration.



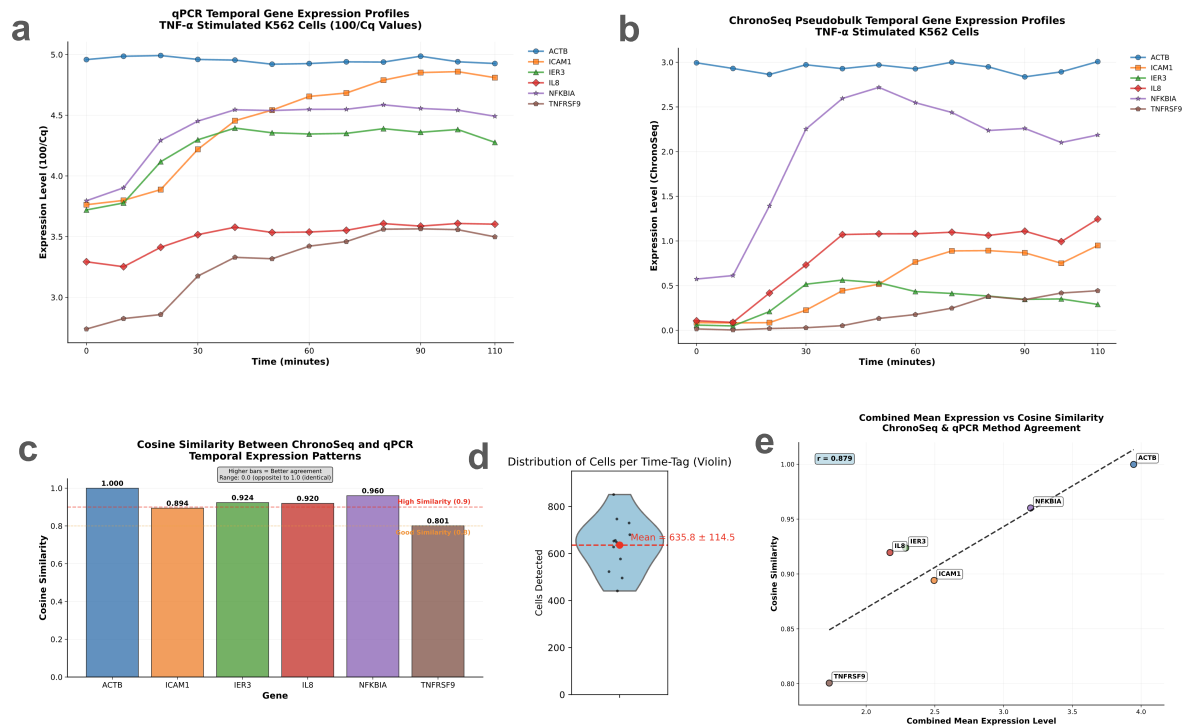
**Supplementary Figure S3: Drop-seq bead modification workflow for Creating ChronoSeq Time-Tagged beads.** Sequences are added by using an oligo that binds to the existing PolyT region on Drop-seq beads. The 3' end of the Drop-seq beads are then extended to the reverse complement of the oligo using E. Coli DNA Polymerase I. Alkaline denaturation is used to make the double stranded DNA single-stranded. These beads are then resuspended in lysis buffer and can be used directly with our device.



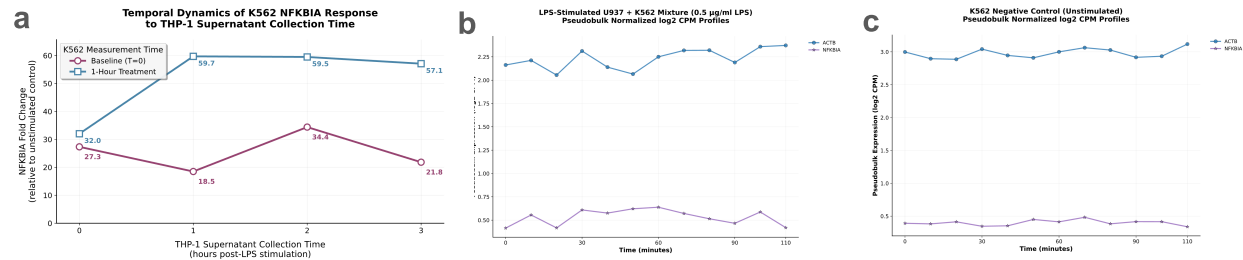
**Supplementary Figure S4: Bulk RNA-Seq Validation of all 12 Time-Tags.** We checked our ability to uniquely identify each Time-Tag. We did this by mixing beads from odd numbered Time-Tags with human cells and even numbered Time-Tags with mouse cells manually in bulk. These beads were then washed and combined for reverse transcription, library preparation and sequencing. Time-Tag resolved cleanly into either human or mouse dominant with 1.6% or less mixed beads detected for each Time-Tag. 10 $\mu$ l of Time-Tagged beads at 450 beads/ $\mu$ l suspended in lysis buffer were directly mixed with 8 $\mu$ l of mouse or human cells at 600cells/ $\mu$ l. Odd numbered Time-Tags were mixed with human(K562) cells while even numbered Time-Tags were mixed with mouse(EL4) cells. Each dot in the scatterplots represents a unique Cell Barcode and the (X, Y) coordinates represent the number of human and mouse transcripts captured respectively. Each plot is also labelled with the which species of the cells were mixed with the beads for the experiment.



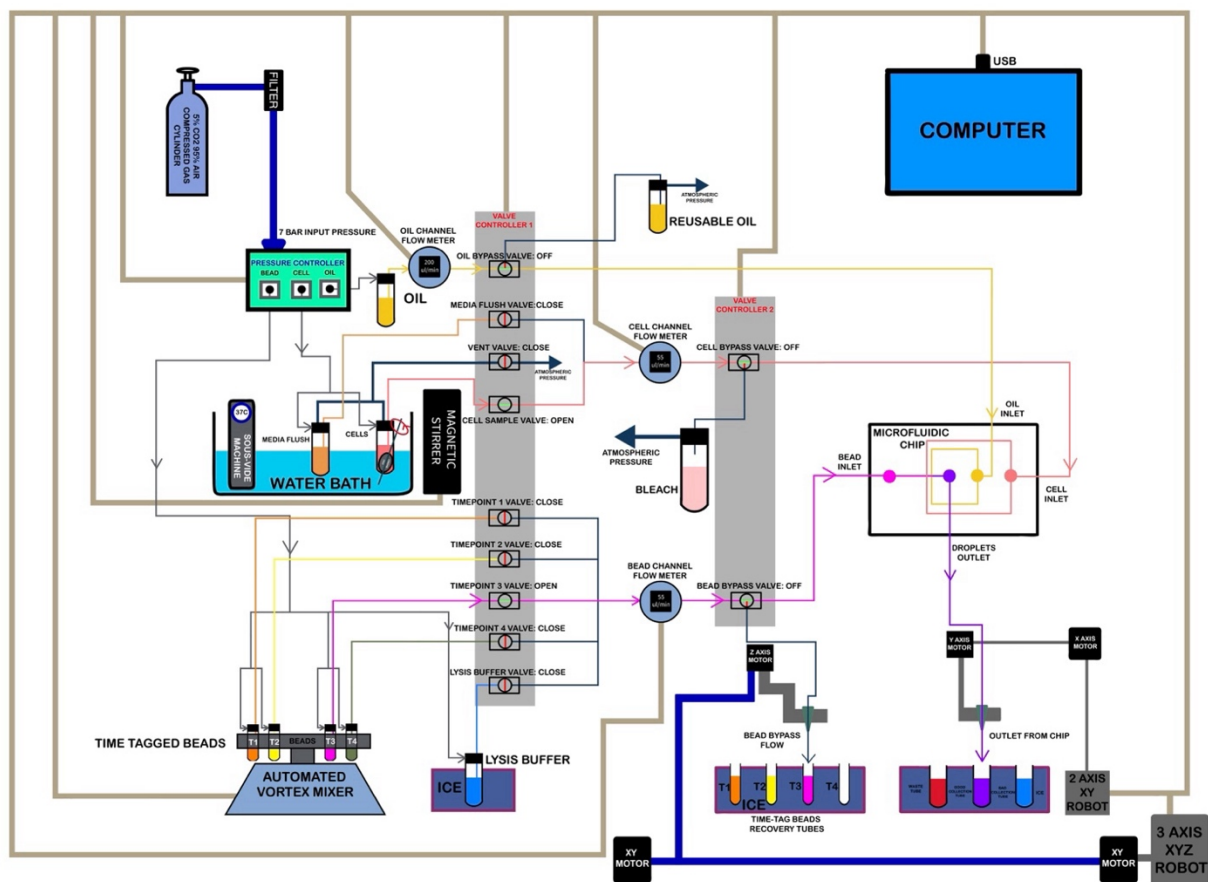
**Supplementary Figure S5: Single-Cell ChronoSeq Experiment to verify unique Time-Tagging:** We checked whether the ChronoSeq device could uniquely tag each sample with a Time-Tag without mixing across samples. To do this we setup three separate reservoirs for maintaining cell suspensions inside the ChronoSeq device. Cell could be sampled from either Reservoir 1, 2 or 3 and co-injected with any of the Time-tagged beads labeled 1-12. A 50:50 mixture of human(K562) and mouse(EL4) cells was added to Reservoir 1, a human only (K562) suspension to Reservoir 2 and a mouse only(EL4) suspension to Reservoir 3. Cells were sampled with a 7-minute sampling interval including a 1-minute sampling duration using the ChronoSeq device. The first sample (left) was from a 50:50 Human(K562) and Mouse(EL4) Cell Suspension. The second sample (center) was from a Human only cell suspension. While the third sample (right) was from a Mouse only cell suspension. Each dot in the three plots represents a Unique Cell Barcode and the (X, Y) coordinates represent the number of Human and Mouse transcripts respectively. Cells were injected at a concentration of 215 cells/ $\mu$ l while the bead concentration was 450 beads/ $\mu$ l. The first, second and third cell suspensions were sequentially co-injected with Time-Tags 10 through 12. Species-mixing barnyard plots clearly showed Time-Tag 10 capturing both human and mouse cells, Time-Tag 11 capturing human cells and Time-Tag 12 capturing mouse cells.



**Supplementary Figure S6:** As a first demonstration of ChronoSeq, we perturbed K562 cells with 10 ng/ml TNF- $\alpha$  at 5 min after the first sample. We sampled every 10 minutes for 12 timepoints to capture the earliest phases of NF- $\kappa$ B signaling. We used a cell concentration of 430 cells/ $\mu$ l and a bead concentration of 450beads/ $\mu$ l for all Time-Tags. To validate the temporal response, we performed a qPCR measurement with the same time resolution of 10 minutes for K562 cells stimulated with TNF- $\alpha$ . We show 5 representative genes and one housekeeping gene (ACTB) and compare it to the pseudobulk of the same genes obtained from the single-cell ChronoSeq data. The temporal profiles (a-b) showed good to high concordance (measured by cosine similarity, c) with higher expressed genes showing higher similarity, thus less affected by noise, compared to lower expressed genes (e). We captured an average of 636 cells per Time-Tag. (d) shows a violin plot of the number of cells we captured per Time-Tag for this experiment.



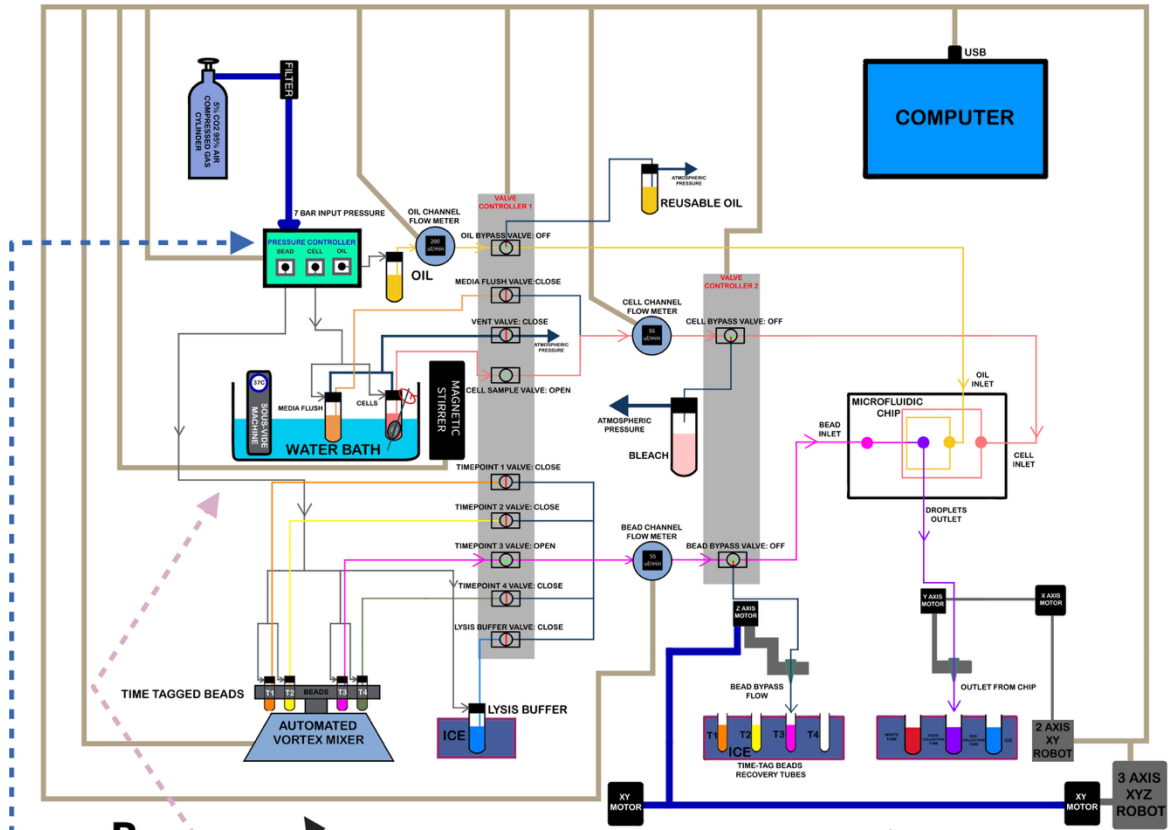
**Supplementary Figure S7:** We validated the lack of responsiveness for K562 cells with a qPCR experiment where supernatants from THP1 cells stimulated with LPS were added to K562 cells at different times after LPS stimulation. NFKBIA was measured as a marker of the inflammatory response in K562 cells and ACTB was measured as a control. K562 supernatants show strong activation for supernatants taken after 1 hour or longer from THP1 cells: (a) NFKBIA fold changes were calculated using the  $2^{-(\Delta\Delta Cq)}$  method<sup>55</sup> with ACTB as the housekeeping gene and unstimulated K562 cells as the reference control. Temporal dynamics were visualized by plotting fold change values against THP-1 supernatant collection time (0, 1, 2, 3 hours post-LPS stimulation). Two K562 measurement timepoints were compared: baseline (T=0, immediately after supernatant addition) and 1-hour post-treatment (T=1h). Data points represent mean values from n=2 technical replicates. Plots clearly show K562 response to stimulation with THP-1 supernatants collected at 1 hour with double the fold change. To further show that K562 cells cannot respond to LPS either alone or when mixed with cells unresponsive to LPS; we LPS stimulated a 50:50 mixture of K562 and undifferentiated U937 cells which are known to show limited or no response to LPS<sup>56</sup>. Unlike the pseudobulks of NFKBIA and ACTB in K562 cells stimulated with TNF- $\alpha$  (**Fig. S6(b)**), the U937:K562 mixture shows no response: (b) Pseudobulks for scRNA-seq generated for ACTB and NFKBIA genes from a 50:50 U937 and K562 cells mixture stimulated with 0.5ug/ml LPS inside the ChronoSeq device. Samples were taken every 10 minutes for 12 timepoints, with LPS addition 5 minutes after first sample. This lack of response is similar to NFKBIA and ACTB pseudobulks from the K562 negative control experiment: (c) Pseudobulks for scRNA-seq generated for ACTB and NFKBIA genes from the K562 negative control experiment.



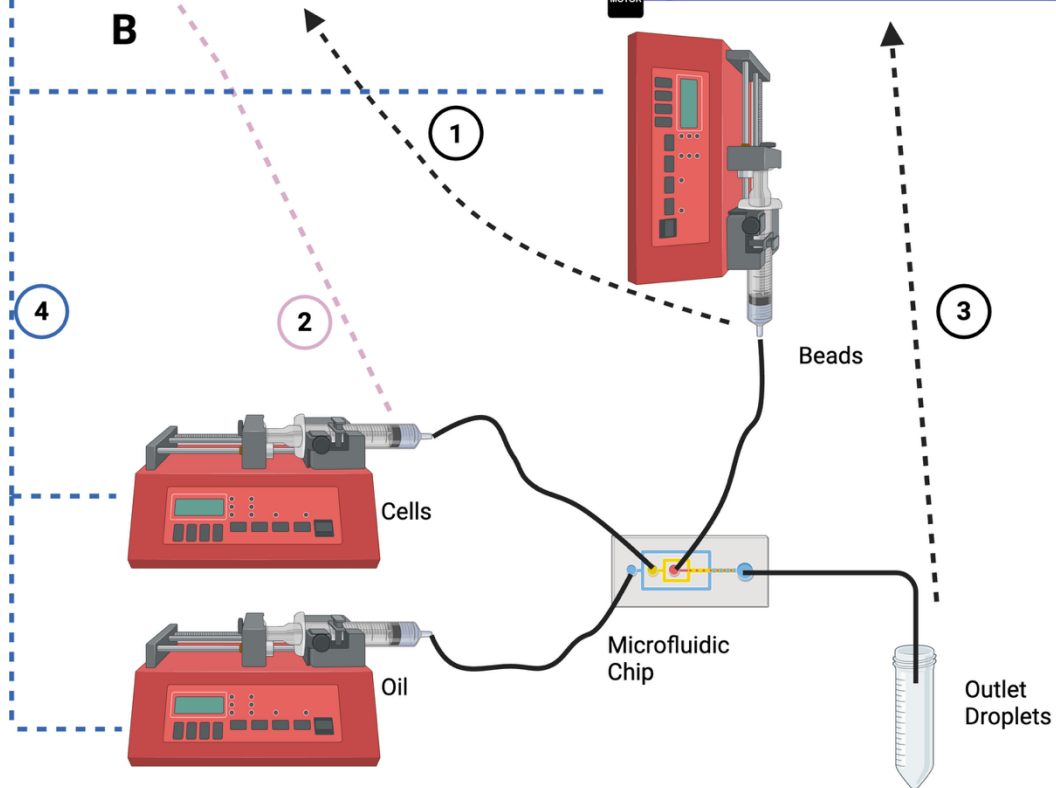
Supplementary Figure S8: ChronoSeq device full schematic



**A**



**B**

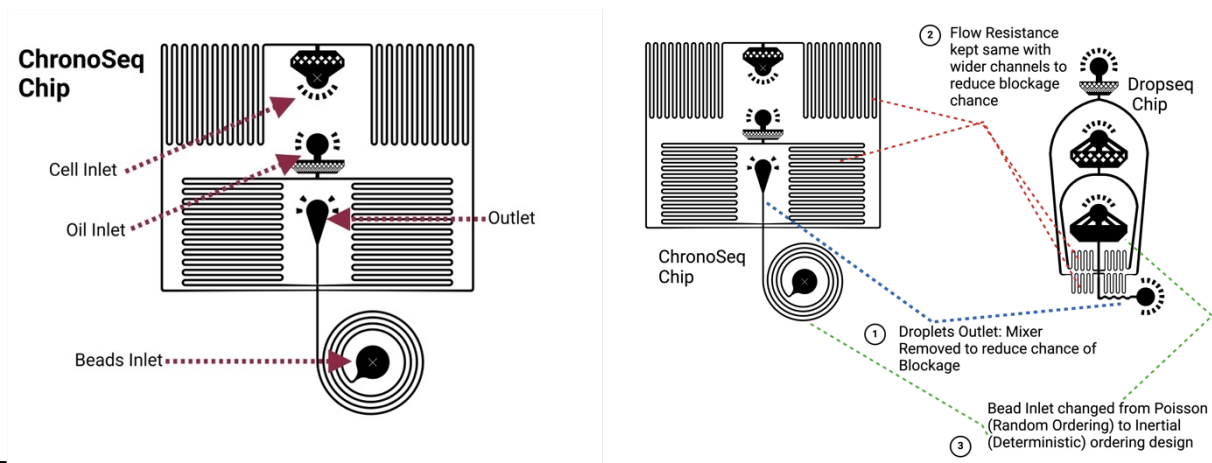


### Supplementary Figure S9: Comparison of ChronoSeq and Drop-seq devices

Our device design addresses several challenges not present in the original Drop-seq<sup>7</sup> system. First, extended cell viability. We maintain cell suspension viability for experiments lasting 2 hours or longer, compared to typical 30-minute Drop-seq runs. Second, cross-contamination prevention. We implemented a washing system to remove unused cells or beads between samples. Third, sample isolation. We designed a system to prevent mixing wash fluid with droplets. To address these challenges, our device operates in two main phases during each injection cycle. 1. Flushing phase that clears fluid lines of residual beads or cells from previous injections. 2. Injection phase that precisely introduces cells and Time-Tagged beads into the microfluidic chip. This integrated system enables controlled, sequential introduction of cells and Time-Tagged beads, capturing gene expression dynamics across multiple time points in a single experiment. Our design prioritizes cell viability maintenance, precise fluid control, and efficient bead usage, all critical for generating high-quality, time-resolved single-cell sequencing data.

The ChronoSeq device is controlled via serial connections to a computer, which coordinates the actions of its various components, while the Drop-seq device works by setting flow rates manually on syringe pumps. Some key features distinguish the ChronoSeq device:

1. **Bead handling:** ChronoSeq incorporates an automated vortex mixer to keep the beads evenly suspended before injection. Drop-seq uses a magnetic stirrer for bead suspension, which leads to high dead volumes and complicates automation for multiple Time-Tags. ChronoSeq allows for the easy addition of more Time-Tags in the future if necessary. Additionally, loading and unloading the beads is simplified; we can directly screw a new 50 mL tube containing beads into the reservoir. The Time-Tagged bead reservoirs are connected to timepoint valves, which facilitate the selection of the appropriate Time-Tagged beads for injection into the microfluidic chip.
2. **Cell culture system:** In the Drop-seq device, cells are directly loaded into a syringe for injection, typically running for 15 to 30 minutes. However, our goal is to maintain cell viability for at least 2 hours. Leaving cells at room temperature without growth media and in the absence of 5% CO<sub>2</sub> can lead to excessive cell death and unwanted perturbations. To address this, we have integrated a cell culture system directly into the ChronoSeq device, featuring a water bath to maintain cells at 37°C, along with a 5% CO<sub>2</sub> gas cylinder that serves as a pressure source and aids in cell culture. Furthermore, we can add additional tubes to culture multiple cell suspensions separately and use valves to select which cell suspension is injected into the microfluidic chip.
3. **Droplet collection:** In the Drop-seq device, droplets are collected directly in a 50 mL tube kept on ice. However, in our system, it is crucial to prevent the beads and cells from previous injections from mixing with those from new injections. To achieve this, we developed a washing system to clear the fluid lines of residual beads and cells. We also need to ensure that washing liquids and unused cell suspensions do not mix with the collected droplets. To solve this issue, we designed a robotic system that facilitates droplet collection and bead recovery. This robot automatically moves the outlet tubing to the correct tube, preventing the mixing of droplets with washing fluids. Additionally, it can recover unused beads from the fluid lines, minimizing waste and reducing experimental costs. Like the Drop-seq device, the droplets in our system are also kept on ice using ice boxes.
4. **Flow control:** In the Drop-seq device, precise speed control is essential for droplet formation. Speeds that are too high can inhibit droplet formation, while low speeds can negatively impact throughput. To achieve this precise control, Drop-seq employs syringe pumps, which allow direct manipulation of the syringe's squeezing rate. In our case, we require high flow rates for flushing and low flow rates for droplet formation. Low flow rates like Drop-seq flow rates, are insufficient for quickly washing unused beads and cells from the device. Therefore, we implemented a compressed air pressure controller and flow sensors for each of the bead, cell, and oil channels. The compressed air controller enables rapid changes in flow rate, while the flow sensors provide the precise control needed for high-quality droplet formation during low flow rate operation.



**Supplementary Figure S10. ChronoSeq microfluidic chip.** The primary function of the microfluidic chip is to co-encapsulate cells with Time-Tagged beads using a flow-focusing junction<sup>57</sup> at the center of the chip. While we initially considered using the Drop-seq device for this purpose, several limitations made it unsuitable for our application: 1. Ordering mechanism. Drop-seq relies on random (Poisson) ordering for co-encapsulation of cells and beads. This random ordering makes it difficult to increase the number of cells collected per timepoint without also increasing droplets containing multiple cells or beads. The ChronoSeq microfluidic chip uses an optimized version of an existing inertial ordering design<sup>58</sup> to increase cell capture through deterministic ordering of beads into droplets. This deterministic ordering allows for capturing more cells by increasing the number of droplets containing exactly one bead and one cell. 2. Channel design. Both the Drop-seq chip and existing inertial ordering<sup>58</sup> designs use narrow flow resistance channels, which are prone to blockage during the flushing phase. To reduce blockage risk, we widened the channels in our chip design. To maintain the same flow resistance with wider channels, we increased the length of the channels to compensate.

**Left**, ChronoSeq chip design with labeled inlet and outlet holes. **Right**, comparison of the chip design. The ChronoSeq chip has three key modifications compared to the Drop-seq chip. 1: The Droplet Mixer near the Drop-seq outlet hole has been removed to reduce the likelihood of blockage. 2: The width of the flow-resistors has been increased along with the length of the channels to keep the flow resistance identical between the two designs. 3: The straight (Poisson) ordering inlet has been replaced with an inertial (deterministic) ordering spiral design. These modifications aim to improve the chip's performance in terms of reduced clogging, maintained flow dynamics, and enhanced cell/bead ordering for more efficient single-cell encapsulation. The inertial ordering design, in particular, allows for a more deterministic approach to cell and bead pairing, potentially increasing the capture efficiency of single cells with single beads.

**Supplementary Table S1: Cq Values for HPRT1 for K562 cells stimulated THP-1 supernatant and controls:** NaN means no Cq value obtained due to lack of amplification for that sample. Results are consistent across both replicates and show addition of THP-1 supernatant leads to RNA degradation.

| Gene  | Stimulant for K562 cells                                         | K562 cells stimulation duration | Replicate | Cq    |
|-------|------------------------------------------------------------------|---------------------------------|-----------|-------|
| HPRT1 | LPS stimulated THP-1 cells supernatant taken at 2 hours          | 1 hour                          | 1         | NaN   |
| HPRT1 | LPS stimulated THP-1 cells supernatant taken at 2 hours          | 1 hour                          | 2         | NaN   |
| HPRT1 | LPS stimulated THP-1 cells supernatant taken immediately-0 hours | 1 hour                          | 1         | NaN   |
| HPRT1 | LPS stimulated THP-1 cells supernatant taken immediately-0 hours | 1 hour                          | 2         | NaN   |
| HPRT1 | Cell culture media DMEM- No stimulation                          | 1 hour                          | 1         | 32.48 |
| HPRT1 | Cell culture media DMEM-No stimulation                           | 1 hour                          | 2         | 32.29 |
| HPRT1 | 10ng/ml TNF- $\alpha$                                            | 1 hour                          | 1         | 31.88 |
| HPRT1 | 10ng/ml TNF- $\alpha$                                            | 1 hour                          | 2         | 31.68 |
| HPRT1 | Blank Control no K562 cells only water                           | N/A                             | 1         | NaN   |
| HPRT1 | Blank Control no K562 cells only water                           | N/A                             | 2         | NaN   |

## REFERENCES

1. Bunne, C. *et al.* How to build the virtual cell with artificial intelligence: Priorities and opportunities. *Cell* **187**, 7045–7063 (2024).
2. Roohani, Y. H. *et al.* Virtual Cell Challenge: Toward a Turing test for the virtual cell. *Cell* **188**, 3370–3374 (2025).
3. Hafner, A. *et al.* p53 pulses lead to distinct patterns of gene expression albeit similar DNA-binding dynamics. *Nat. Struct. Mol. Biol.* **24**, 840–847 (2017).
4. Zambrano, S., De Toma, I., Piffer, A., Bianchi, M. E. & Agresti, A. NF- $\kappa$ B oscillations translate into functionally related patterns of gene expression. *eLife* **5**, e09100 (2016).
5. Lane, K. *et al.* Measuring Signaling and RNA-Seq in the Same Cell Links Gene Expression to Dynamic Patterns of NF- $\kappa$ B Activation. *Cell Syst.* **4**, 458-469.e5 (2017).
6. Chen, W. *et al.* Live-seq enables temporal transcriptomic recording of single cells. *Nature* **608**, 733–740 (2022).
7. Macosko, E. Z. *et al.* Highly Parallel Genome-wide Expression Profiling of Individual Cells Using Nanoliter Droplets. *Cell* **161**, 1202–1214 (2015).
8. Klein, A. M. *et al.* Droplet Barcoding for Single-Cell Transcriptomics Applied to Embryonic Stem Cells. *Cell* **161**, 1187–1201 (2015).
9. Han, X. *et al.* Mapping the Mouse Cell Atlas by Microwell-Seq. *Cell* **172**, 1091-1107.e17 (2018).
10. Shalek, A. K. *et al.* Single-cell RNA-seq reveals dynamic paracrine control of cellular variation. *Nature* **510**, 363–369 (2014).
11. Zheng, G. X. Y. *et al.* Massively parallel digital transcriptional profiling of single cells. *Nat. Commun.* **8**, 14049 (2017).

12. Rosenberg, A. B. *et al.* Single-cell profiling of the developing mouse brain and spinal cord with split-pool barcoding. *Science* **360**, 176–182 (2018).
13. Lake, B. B. *et al.* Neuronal subtypes and diversity revealed by single-nucleus RNA sequencing of the human brain. *Science* **352**, 1586–1590 (2016).
14. Clark, I. C. *et al.* *Microfluidics-Free Single-Cell Genomics with Templated Emulsification*. (2022) doi:10.1101/2022.06.10.495582.
15. Love, J. C., Gierahn, T. M., Shalek, A. K., Wadsworth, M. H. & Hughes, T. K. Single cell analyses. (2024).
16. Gierahn, T. M. *et al.* Seq-Well: portable, low-cost RNA sequencing of single cells at high throughput. *Nat. Methods* **14**, 395–398 (2017).
17. Tay, S. *et al.* Single-cell NF- $\kappa$ B dynamics reveal digital activation and analogue information processing. *Nature* **466**, 267–271 (2010).
18. Qiu, Q. *et al.* Massively parallel and time-resolved RNA sequencing in single cells with scNT-seq. *Nat. Methods* **17**, 991–1001 (2020).
19. Davari, K. *et al.* Rapid Genome-wide Recruitment of RNA Polymerase II Drives Transcription, Splicing, and Translation Events during T Cell Responses. *Cell Rep.* **19**, 643–654 (2017).
20. Erhard, F. *et al.* scSLAM-seq reveals core features of transcription dynamics in single cells. *Nature* **571**, 419–423 (2019).
21. Battich, N. *et al.* Sequencing metabolically labeled transcripts in single cells reveals mRNA turnover strategies. *Science* **367**, 1151–1156 (2020).
22. Erhard, F. *et al.* Time-resolved single-cell RNA-seq using metabolic RNA labelling. *Nat. Rev. Methods Primer* **2**, 1–18 (2022).

23. Liu, H. *et al.* SLAM-Drop-seq reveals mRNA kinetic rates throughout the cell cycle. *Mol. Syst. Biol.* **19**, e11427 (2023).
24. Mahat, D. B. *et al.* Single-cell nascent RNA sequencing unveils coordinated global transcription. *Nature* **631**, 216–223 (2024).
25. Purvis, J. E. & Lahav, G. Encoding and Decoding Cellular Information through Signaling Dynamics. *Cell* **152**, 945–956 (2013).
26. Gómez-Sjöberg, R., Leyrat, A. A., Pirone, D. M., Chen, C. S. & Quake, S. R. Versatile, Fully Automated, Microfluidic Cell Culture System. *Anal. Chem.* **79**, 8557–8563 (2007).
27. Kellogg, R. A., Gómez-Sjöberg, R., Leyrat, A. A. & Tay, S. High-throughput microfluidic single-cell analysis pipeline for studies of signaling dynamics. *Nat. Protoc.* **9**, 1713–1726 (2014).
28. Yang, J.-M. *et al.* Deciphering cell signaling networks with massively multiplexed biosensor barcoding. *Cell* **184**, 6193–6206.e14 (2021).
29. Zhao, S. *et al.* A single-cell massively parallel reporter assay detects cell-type-specific gene regulation. *Nat. Genet.* **55**, 346–354 (2023).
30. Schiebinger, G. *et al.* Optimal-Transport Analysis of Single-Cell Gene Expression Identifies Developmental Trajectories in Reprogramming. *Cell* **176**, 928–943.e22 (2019).
31. Van den Berge, K. *et al.* Trajectory-based differential expression analysis for single-cell sequencing data. *Nat. Commun.* **11**, 1201 (2020).
32. Guo, Q. *et al.* NF- $\kappa$ B in biology and targeted therapy: new insights and translational implications. *Signal Transduct. Target. Ther.* **9**, 53 (2024).
33. Türei, D. *et al.* Integrated intra- and intercellular signaling knowledge for multicellular omics analysis. *Mol. Syst. Biol.* **17**, e9923 (2021).

34. Moerman, T. *et al.* GRNBoost2 and Arboreto: efficient and scalable inference of gene regulatory networks. *Bioinforma. Oxf. Engl.* **35**, 2159–2161 (2019).
35. Kim, S. ppcor: An R Package for a Fast Calculation to Semi-partial Correlation Coefficients. *Commun. Stat. Appl. Methods* **22**, 665–674 (2015).
36. Papili Gao, N., Ud-Dean, S. M. M., Gandrillon, O. & Gunawan, R. SINCERITIES: inferring gene regulatory networks from time-stamped single cell transcriptional expression profiles. *Bioinforma. Oxf. Engl.* **34**, 258–266 (2018).
37. Huynh-Thu, V. A., Irrthum, A., Wehenkel, L. & Geurts, P. Inferring Regulatory Networks from Expression Data Using Tree-Based Methods. *PLOS ONE* **5**, e12776 (2010).
38. Dunham, I. *et al.* An integrated encyclopedia of DNA elements in the human genome. *Nature* **489**, 57–74 (2012).
39. Chevalley, M., Roohani, Y. H., Mehrjou, A., Leskovec, J. & Schwab, P. A large-scale benchmark for network inference from single-cell perturbation data. *Commun. Biol.* **8**, 412 (2025).
40. Han, H. *et al.* TRRUST v2: an expanded reference database of human and mouse transcriptional regulatory interactions. <https://dx.doi.org/10.1093/nar/gkx1013>.
41. Perik-Zavodskaia, O. *et al.* Knockouts of TNFRSF1A and TNFRSF1B Genes in K562 Cell Line Lead to Diverse Long-Lasting Responses to TNF- $\alpha$ . *Int. J. Mol. Sci.* **24**, 17169 (2023).
42. Takashiba, S. *et al.* Differentiation of Monocytes to Macrophages Primes Cells for Lipopolysaccharide Stimulation via Accumulation of Cytoplasmic Nuclear Factor  $\kappa$ B. *Infect. Immun.* **67**, 5573–5578 (1999).



43. Caldwell, A. B., Cheng, Z., Vargas, J. D., Birnbaum, H. A. & Hoffmann, A. Network dynamics determine the autocrine and paracrine signaling functions of TNF. *Genes Dev.* **28**, 2120–2133 (2014).
44. Jin, S. *et al.* Inference and analysis of cell-cell communication using CellChat. *Nat. Commun.* **12**, 1088 (2021).
45. Granger, C. W. J. Investigating Causal Relations by Econometric Models and Cross-spectral Methods. *Econometrica* **37**, 424–438 (1969).
46. Ubanako, P., Xelwa, N. & Ntwasa, M. LPS induces inflammatory chemokines via TLR-4 signalling and enhances the Warburg Effect in THP-1 cells. *PloS One* **14**, e0222614 (2019).
47. Wan, J. *et al.* NF- $\kappa$ B inhibition attenuates LPS-induced TLR4 activation in monocyte cells. *Mol. Med. Rep.* **14**, 4505–4510 (2016).
48. Mazutis, L. *et al.* Single-cell analysis and sorting using droplet-based microfluidics. *Nat. Protoc.* **8**, 870–891 (2013).
49. Wolf, F. A., Angerer, P. & Theis, F. J. SCANPY: large-scale single-cell gene expression data analysis. *Genome Biol.* **19**, 15 (2018).
50. Wolock, S. L., Lopez, R. & Klein, A. M. Scrublet: Computational Identification of Cell Doublets in Single-Cell Transcriptomic Data. *Cell Syst.* **8**, 281-291.e9 (2019).
51. Kowalczyk, M. S. *et al.* Single-cell RNA-seq reveals changes in cell cycle and differentiation programs upon aging of hematopoietic stem cells. *Genome Res.* **25**, 1860–1872 (2015).
52. Zhang, Y., Parmigiani, G. & Johnson, W. E. ComBat-seq: batch effect adjustment for RNA-seq count data. *NAR Genomics Bioinforma.* **2**, lqaa078 (2020).

53. Sheu, K. M., Pimplaskar, A. & Hoffmann, A. Single-cell stimulus-response gene expression trajectories reveal the stimulus specificities of dynamic responses by single macrophages. *Mol. Cell* **84**, 4095-4110.e6 (2024).
54. Lange, M. *et al.* CellRank for directed single-cell fate mapping. *Nat. Methods* **19**, 159–170 (2022).
55. Livak, K. J. & Schmittgen, T. D. Analysis of relative gene expression data using real-time quantitative PCR and the 2(-Delta Delta C(T)) Method. *Methods San Diego Calif* **25**, 402–408 (2001).
56. Ciubotaru, I., Potempa, L. A. & Wander, R. C. Production of modified C-reactive protein in U937-derived macrophages. *Exp. Biol. Med. Maywood NJ* **230**, 762–770 (2005).
57. Ward, T., Faivre, M., Abkarian, M. & Stone, H. A. Microfluidic flow focusing: Drop size and scaling in pressure versus flow-rate-driven pumping. *ELECTROPHORESIS* **26**, 3716–3724 (2005).
58. Moon, H.-S. *et al.* Inertial-ordering-assisted droplet microfluidics for high-throughput single-cell RNA-sequencing. *Lab. Chip* **18**, 775–784 (2018).